第十讲：目标检测

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知识回顾：图像识别

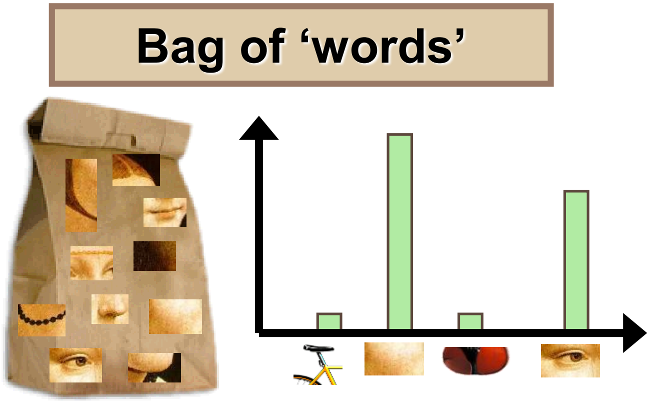
机器学习与图像分类

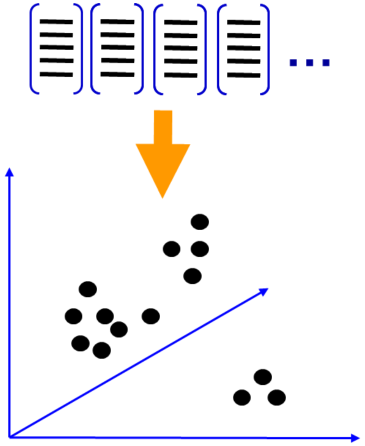
- 有监督学习与无监督学习
- 欠拟合与过拟合
- 基于词袋模型的图像分类

y=f(x,w)

$$w^* = \arg\min_{w} \sum_{i} L(y^i, f(x^i, w))$$

最优参数　　　　损失函数





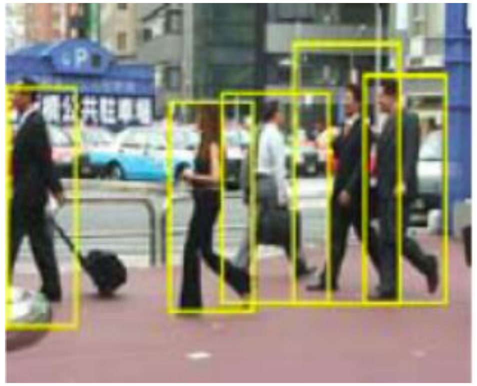
Content

- Object detection basics 目标检测基础
  - What is object detection
  - General process
  - Evaluation
- Pedestrian detection 行人检测
  - Dalal-Triggs method
  - Histogram of oriented gradients
  - Learning with SVM
- Face detection 人脸检测
  - Viola-Jones
  - Boosting for learning

Object detection 目标检测

Object (category) detection: Find the bounding box of object instances in the image

- Find all the people
- Find all the faces
- Often within a single image
- Often 'sliding window'



给出目标物体的边界框

Image classification

- Scenes have "stuff" – distribution of materials and surfaces with arbitrary shape.
- Bag of Features ok!

Object detection

- Objects are "things" with shape, boundaries.
- Bag of Features less ok as spatial layout is lost!

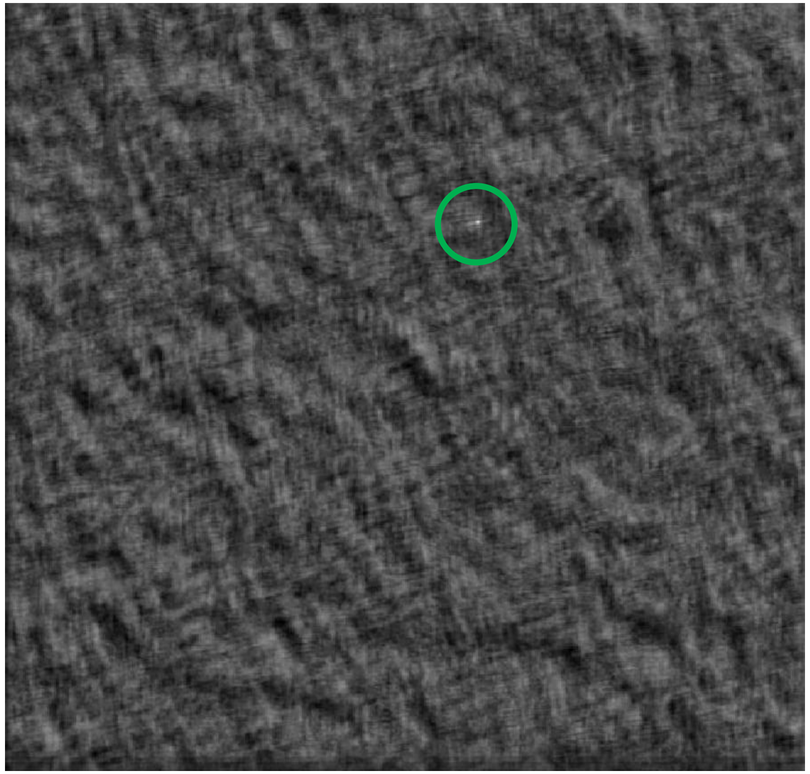
特征袋模型不适合目标检测任务

（无空间布局信息）

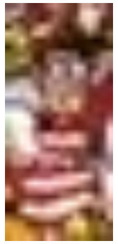
Object detection 目标检测

- A basic method: template matching 模板匹配

Scene



Correlation map



Template

Object category detection

模板匹配

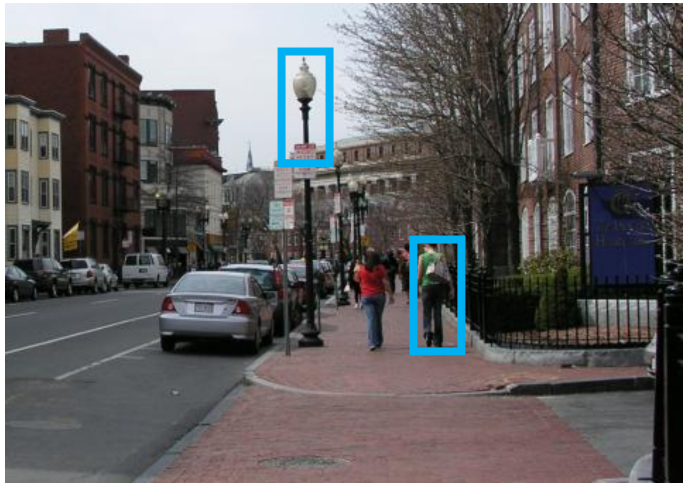
A basic method: template matching

1, Align 空间对齐

- E.g., choose position, scale orientation
- How to make this tractable?

2. Compare 计算相似性

- Compute similarity to an example object or to a summary representation
- Which differences in appearance are important?



Aligned Possible Objects

Exemplar Summary

模板

General Process of Object Recognition 基本流程

- A general object detection method has the following steps

指定目标模型 — **Specify Object Model** — What are the object parameters?

产生假设 — **Generate Hypotheses** — Propose an alignment of the model to the image

假设打分 — **Score Hypotheses** — Calculate the likelihood of the candidate region being the object

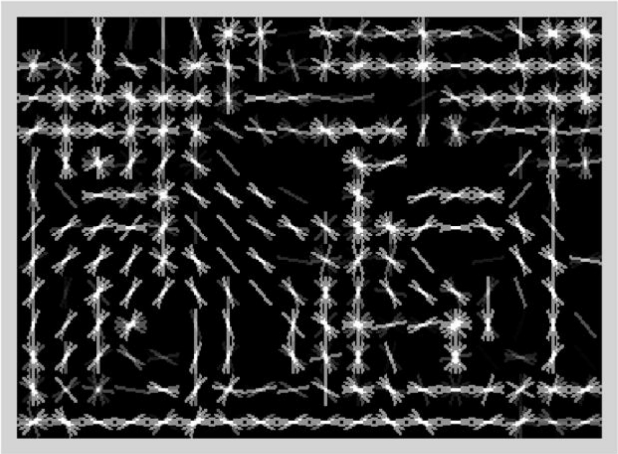
检测解析 — **Resolve Detections**

Specifying an object model 指定目标模型

1. Statistical Template in Bounding Box 统计模型

- Object is some (x,y,w,h) in image
  - (x,y) is the position, w and h are width and height, respectively
- Features defined wrt. bounding box coordinates

Image



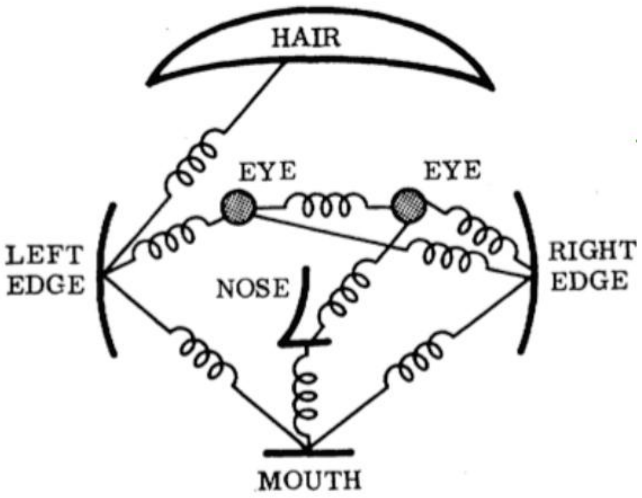
Template Visualization

Each local region is represented by a feature descriptor such as SIFT

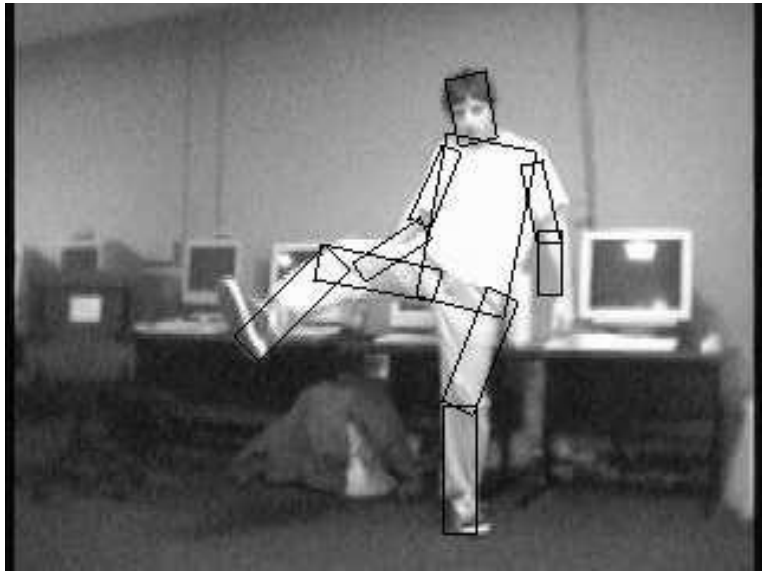
Specifying an object model

2. Articulated parts model 关节部件模型

- Object is configuration of parts
- Each part is detectable



A face model

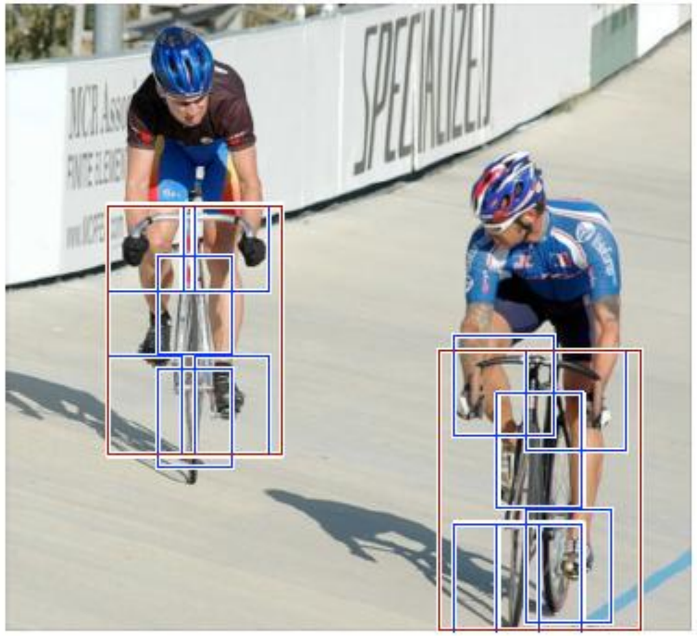


A human model

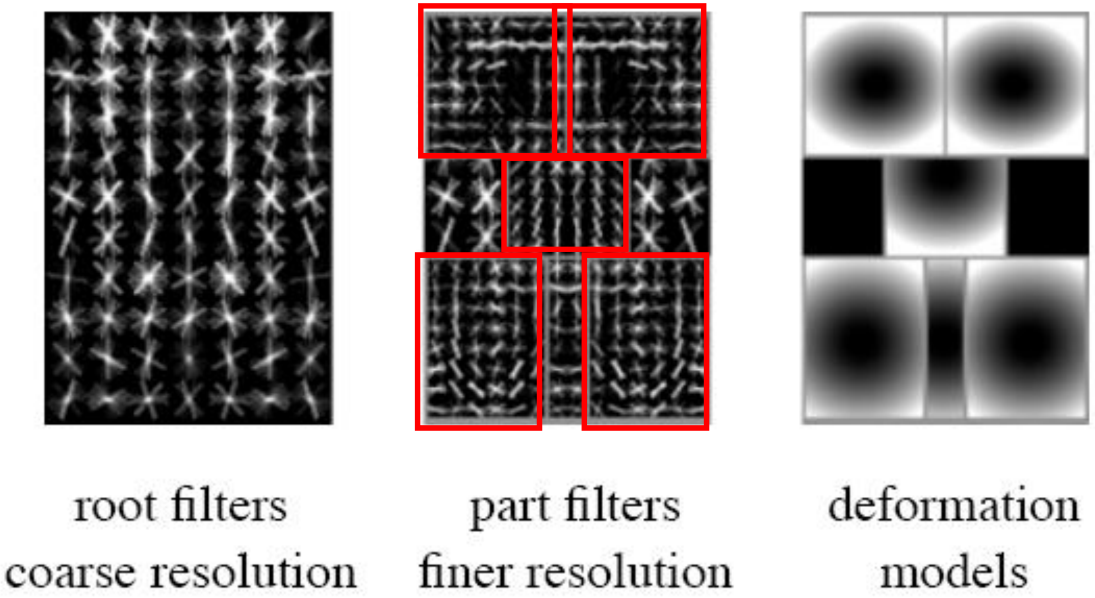
Specifying an object model

3. Hybrid template/parts model 混合模型

- One global model at a coarse resolution
- Parts model on a finer resolution



Detections

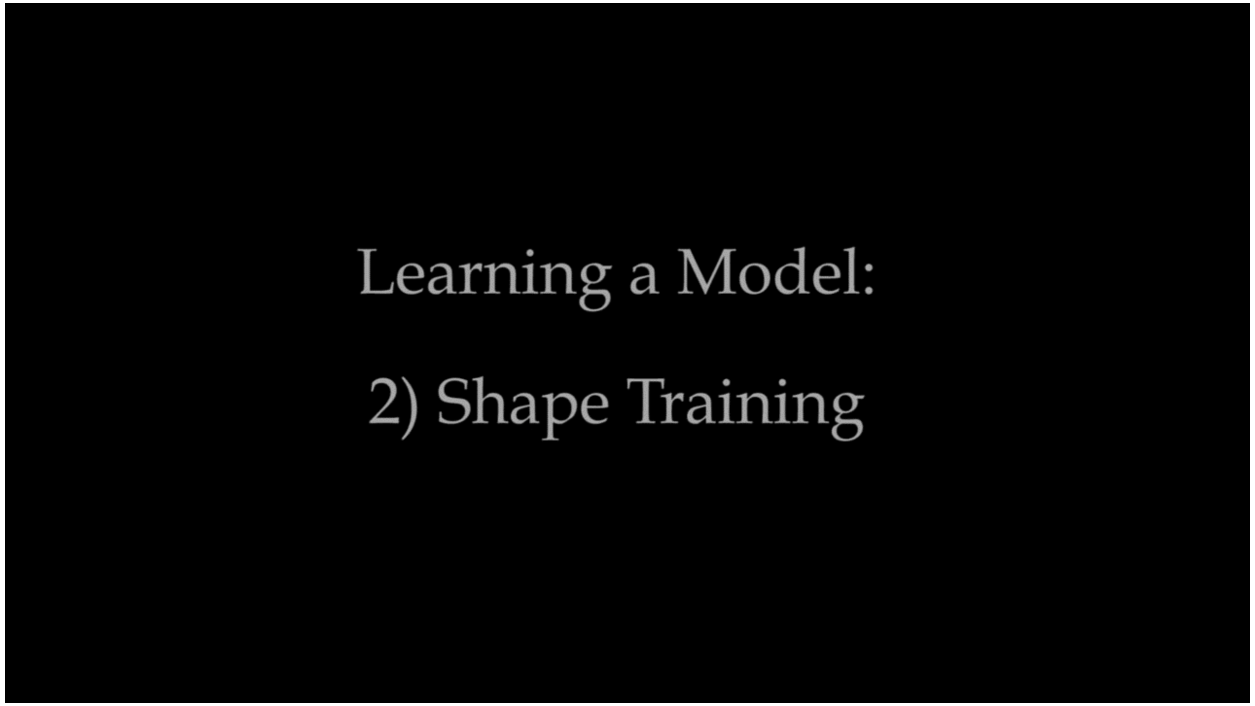


Template Visualization

Specifying an object model

4. Deformable 3D model 可形变三维模型

- Object is a parameterized space of shape/pose/deformation of class of 3D object



SMPL is a realistic 3D model of the human body that is based on skinning and blend shapes and is learned from thousands of 3D body scans.

http://smpl.is.tue.mpg.de/

General Process of Object Recognition

- A general object detection method has the following steps

指定目标模型 — **Specify Object Model** — What are the object parameters?

↓

产生假设 — **Generate Hypotheses** — Propose an alignment of the model to the image

↓

假设打分 — **Score Hypotheses** — Calculate the likelihood of the candidate region being the object

↓

检测解析 — **Resolve Detections**

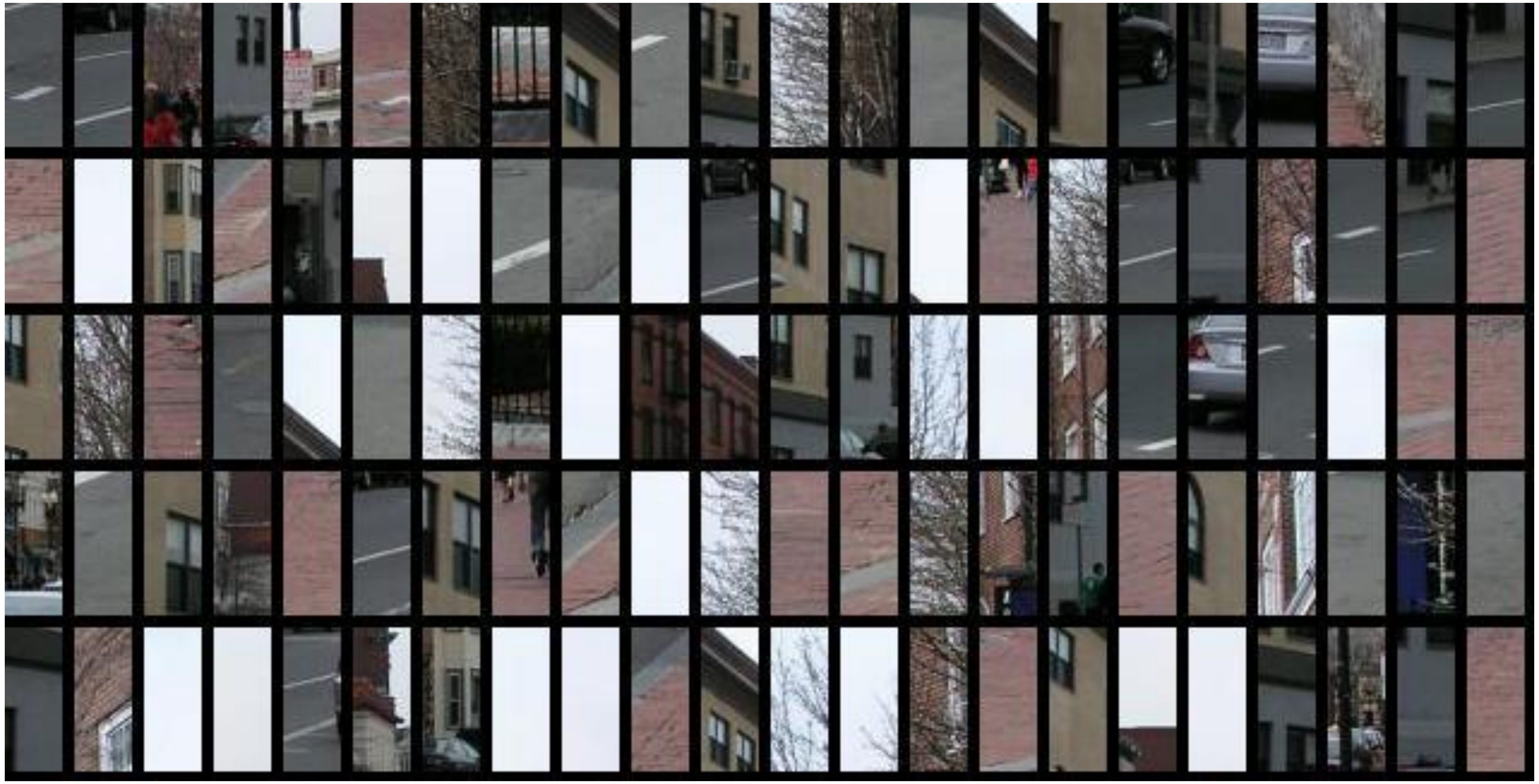
Generating hypotheses 产生假设

1. 2D template model / sliding window 滑动窗口
    - Test patch at each location and scale

Question: how to deal with objects at different scales?
Answer: Resize the image to multiple resolution levels to create an image pyramid, and slide the template across these images respectively.

Generating hypotheses产生假设

1. 2D template model / sliding window 滑动窗口
   - Test patch at each location and scale



Generating hypotheses 产生假设

2. Region-based proposal 区域提议（基于分割）

- Arbitrary bounding box + image ‘cut’ segmentation

Generating hypotheses 产生假设

2. Region-based proposal 区域提议(基于聚类)

- Selective Search by Hierarchical Grouping

通过分层次聚类，产生不同尺度的区域提议

Algorithm 1: Hierarchical Grouping Algorithm

DontPrintSemicolon **Input**: (colour) image
Output: Set of object location hypotheses L

Obtain initial regions $R = \{r_1, \cdots, r_n\}$ using Felzenszwalb and Huttenlocher (2004) Initialise similarity set $S = \emptyset$;
foreach *Neighbouring region pair* (r_i, r_j) **do**
  Calculate similarity $s(r_i, r_j)$;
  $S = S \cup s(r_i, r_j)$;

while $S \neq \emptyset$ **do**
  Get highest similarity $s(r_i, r_j) = \max(S)$;
  Merge corresponding regions $r_t = r_i \cup r_j$;
  Remove similarities regarding r_i : $S = S \setminus s(r_i, r_*)$;
  Remove similarities regarding r_j : $S = S \setminus s(r_*, r_j)$;
  Calculate similarity set S_t between r_t and its neighbours;
  $S = S \cup S_t$;
  $R = R \cup r_t$;
Extract object location boxes L from all regions in R;

General Process of Object Recognition

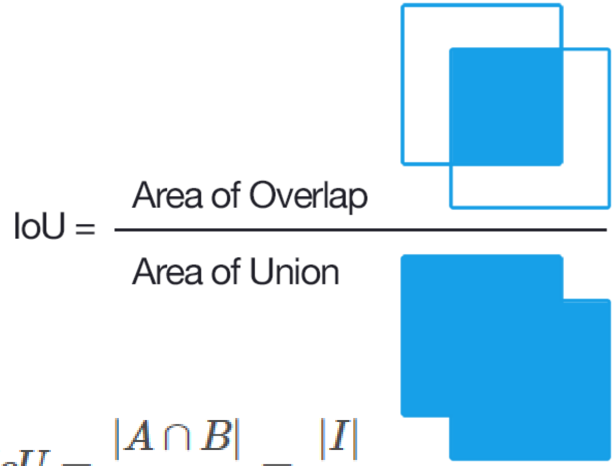
- A general object detection method has the following steps

指定目标模型	Specify Object Model	What are the object parameters?
产生假设	Generate Hypotheses	Propose an alignment of the model to the image
假设打分	Score Hypotheses	Calculate the likelihood of the candidate region being the object， based on summary representation and classifiers
检测解析	Resolve Detections	Rescore each proposed object based on whole set

General Process of Object Recognition

- Resolve Detections 检测解析：非最大值抑制
  - Non-max suppression to filter the proposals

Intersection over union 交并比

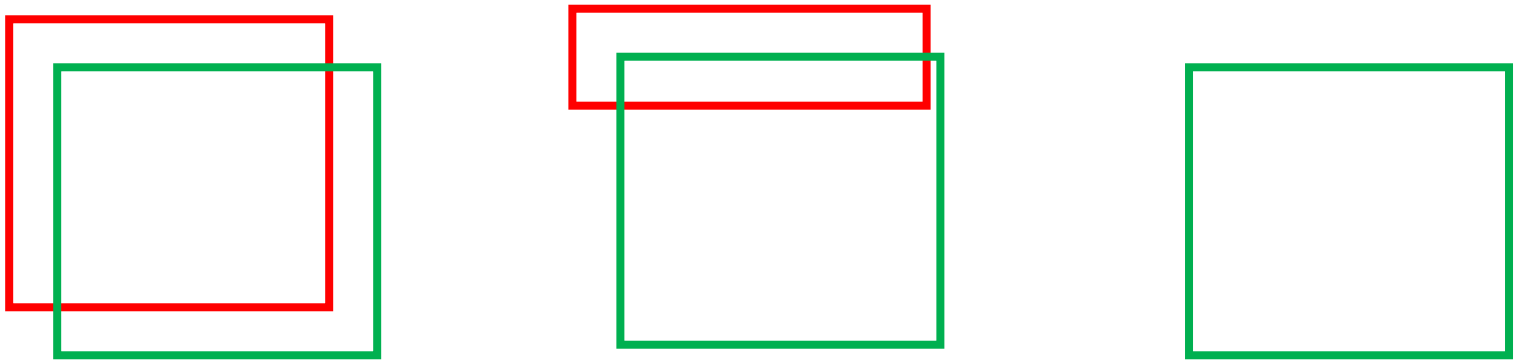


$$IoU = \frac{|A \cap B|}{|A \cup B|} = \frac{|I|}{|U|}$$

Evaluation metrics for detection 检测结果评价

- True positive and false positive

真阳性　　　　假阳性



IoU >= t

True positive

IoU < t

False positive

False negative

Green = ground truth

Red = prediction

Evaluation metrics for detection检测结果评价

- Precision and recall

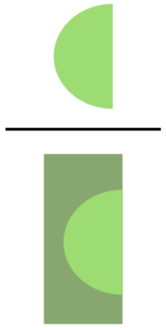
精确度

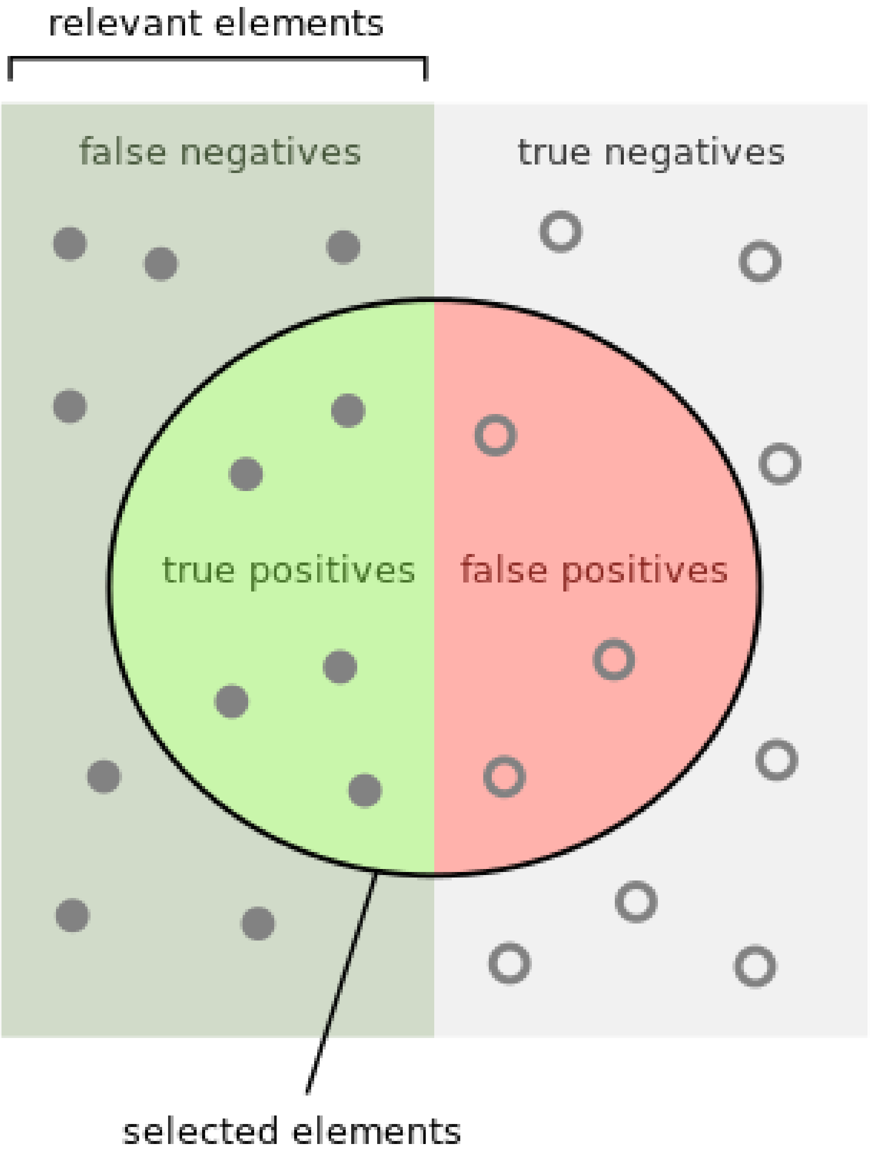
How many selected items are relevant ?

Precision = 

召回率

How many relevant items are selected ?

Recall = 



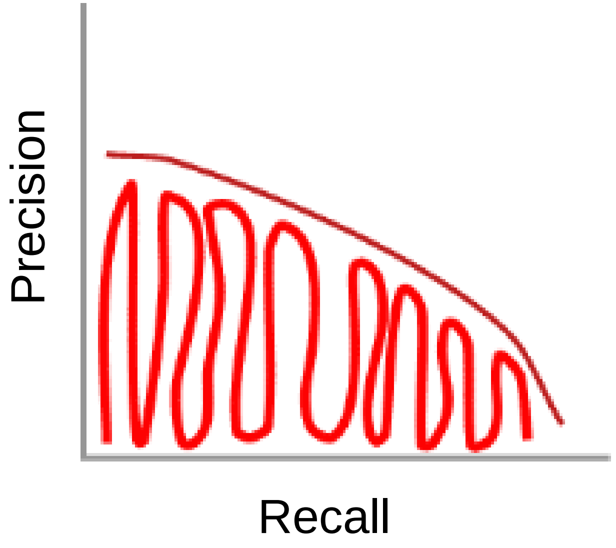
Object Detection

Evaluation Metrics: mAP

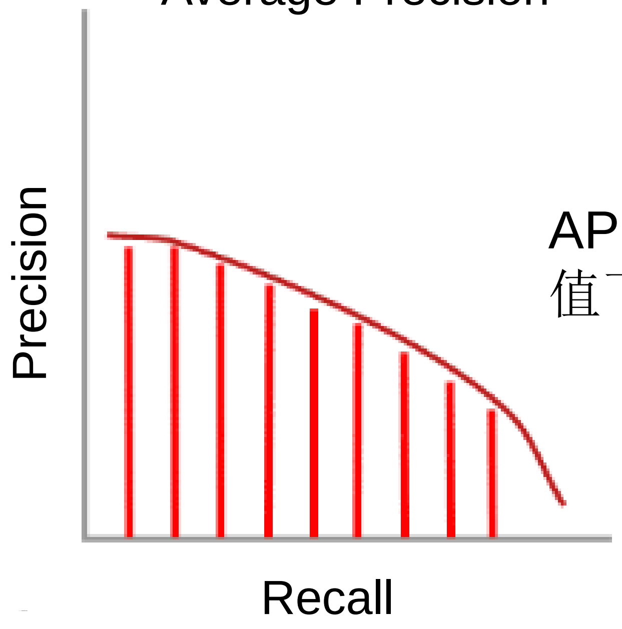
AP: average precision
Consider the precision-recall curve

考虑不同分类阈值情况下所有的Precession和Recall的组合，得到一个单一的评价指标：AUC或AP

AUC
(Area under the curve)



Average Precision



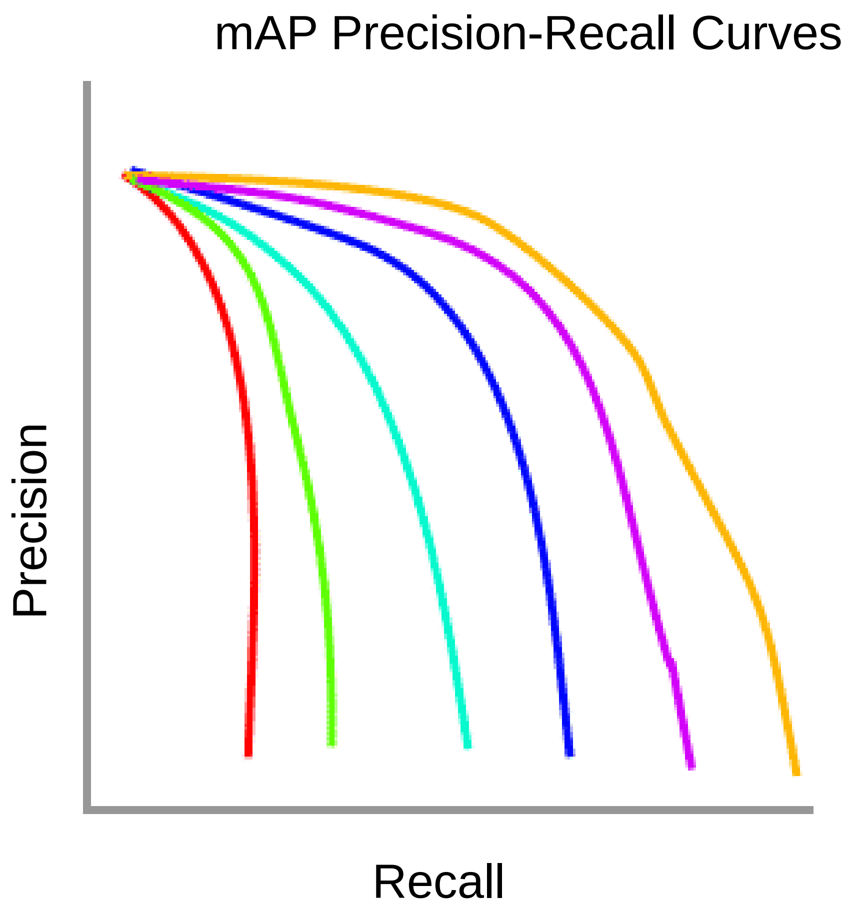
AP = 不同分类阈值对应的Recall值下的Precision的加权平均

Object Detection

Evaluation Metrics: mAP

mAP: mean AP
A series of precision recall curves with the IoU threshold set at varying levels

考虑不同IoU阈值的情况下，所有的Precision-Recall曲线对应的AP的平均值

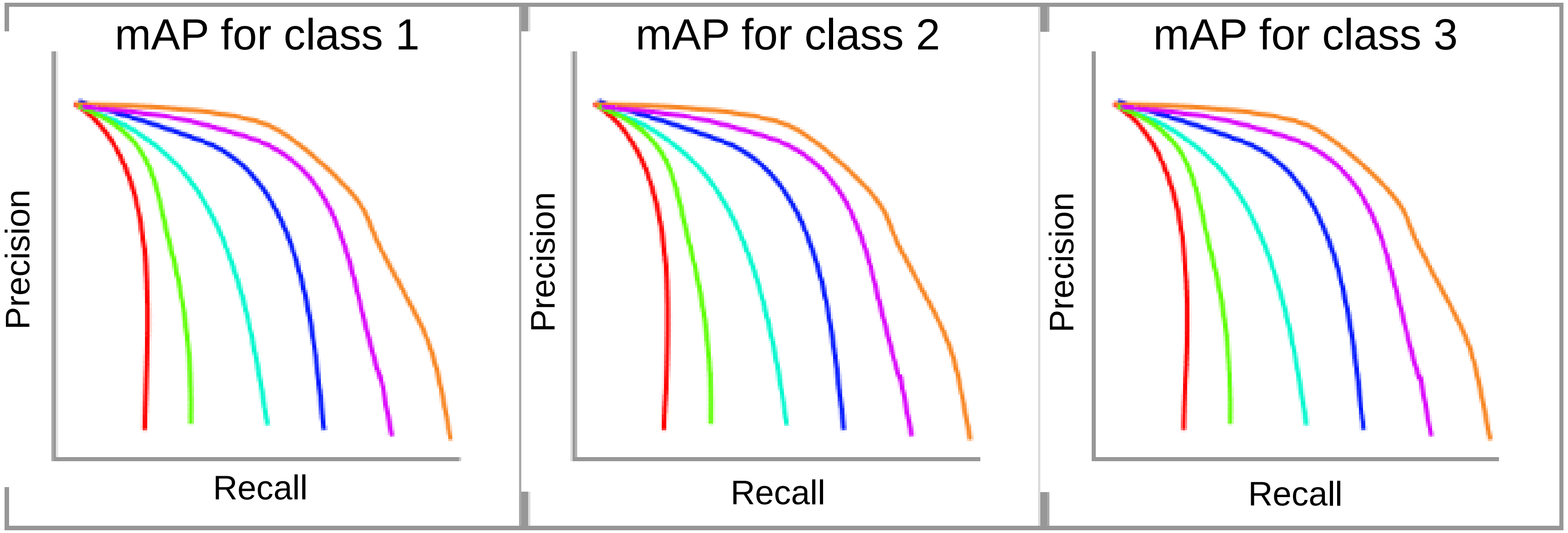


Object Detection

Evaluation Metrics: mAP

mAP: mean AP
A series of precision recall curves with the IoU threshold set at varying levels

针对多类检测问题，再取所有类别的mAP的平均



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Dalal Triggs: Person detection with HOG & linear SVM

Histograms of oriented gradients for human detection
N Dalal, B Triggs - 2005 IEEE computer society conference on ..., 2005 - ieeexplore.ieee.org
... The first need is a robust feature set that allows the human form to be discriminated cleanly,
... sets for human detection, showing that locally normalized Histogram of Oriented Gradient (...
☆ 保存 99 引用 被引用次数：43739 相关文章 所有 87 个版本

Navneet Dalal, Bill Triggs . Histograms of Oriented Gradients for Human Detection, CVPR, 2005

>43k citations

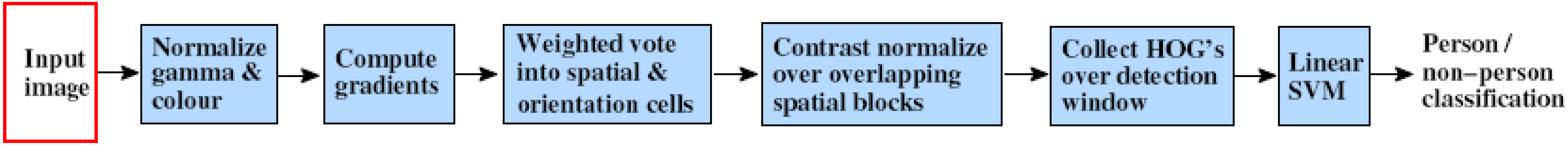
Dalal-Triggs pedestrian detector

1. Extract fixed-sized (64x128 pixel) window at each position and scale
2. Compute HOG (histogram of gradient) features within each window
3. Score the window with a linear SVM classifier
4. Perform non-maxima suppression to remove overlapping detections with lower scores

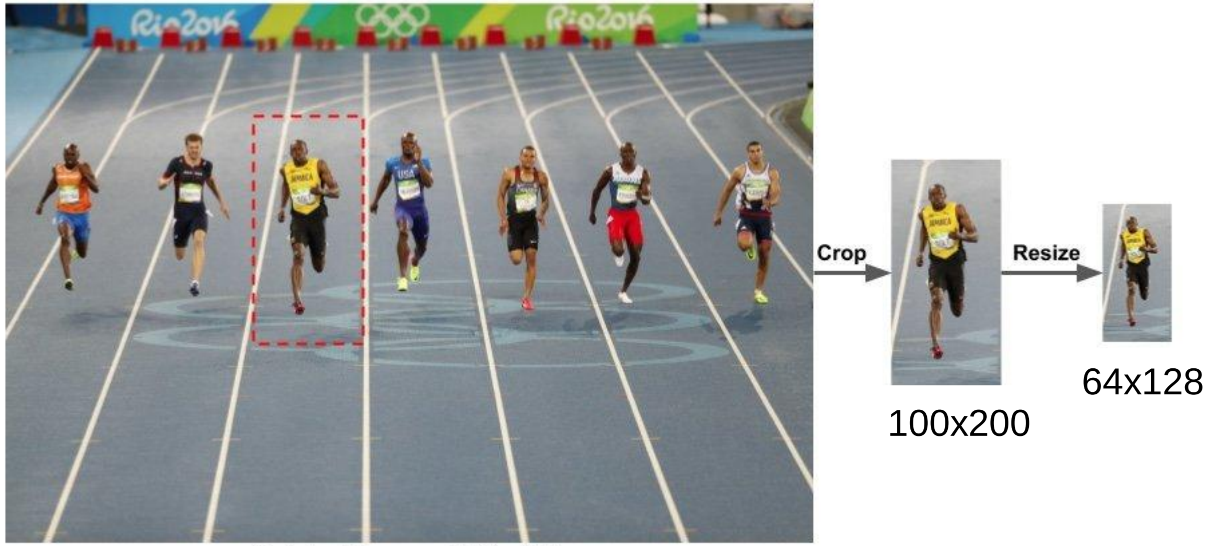
64x128

Dalal-Triggs pedestrian detector

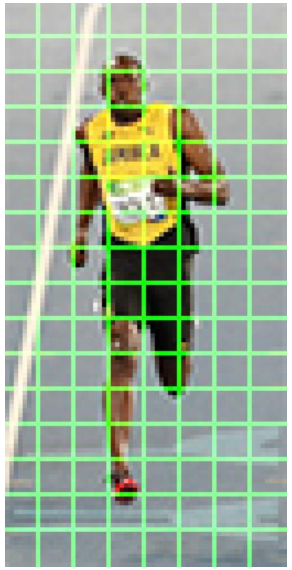
Preprocess



Crop and resize



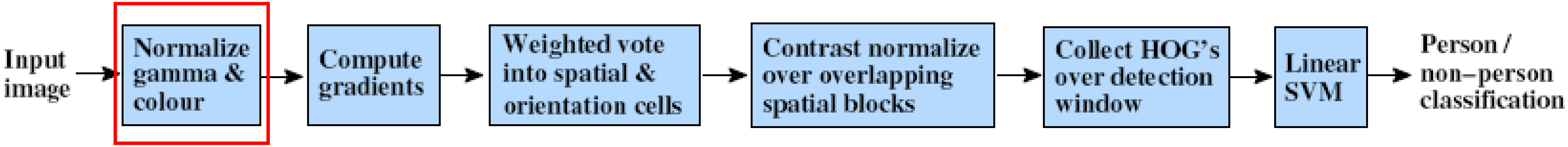
Original image

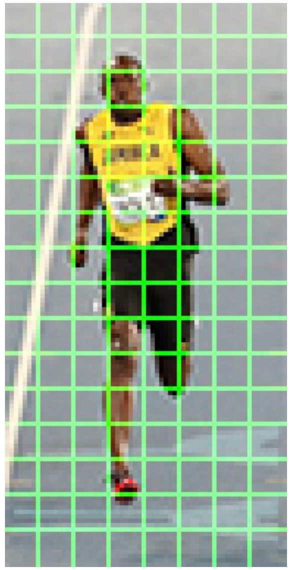


64x128

Dalal-Triggs pedestrian detector

Preprocess





Tested with

- RGB
- LAB
- Grayscale

RGB and LAB: Slightly better performance vs. grayscale

对颜色、光照不敏感

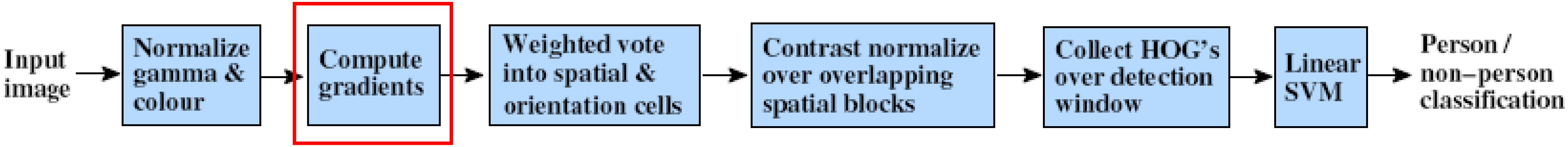
Gamma Normalization and Compression

- Square root
- Log

Square root and Log: Very slightly better performance vs. no adjustment

Dalal-Triggs pedestrian detector

HOG Feature extraction



Outperforms

-1	0	1

centered

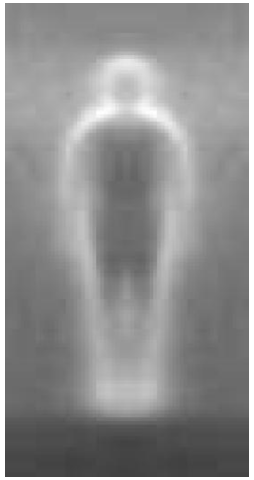
-1	1

uncentered

1	-8	0	8	-1

cubic-corrected

Edge detection



0	1
-1	0

diagonal

-1	0	1
-2	0	2
-1	0	1

Sobel

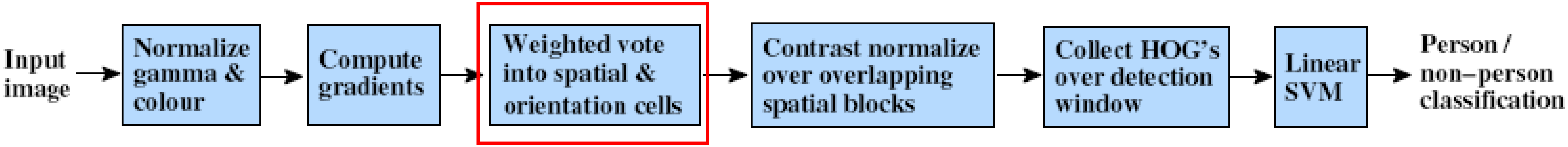
Gx

Gy

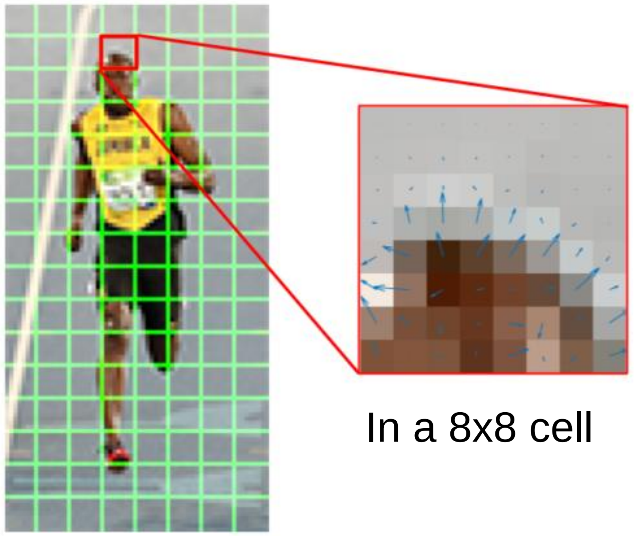
magnitude

Dalal-Triggs pedestrian detector

HOG Feature extraction



Histogram of oriented Gradient



In a 8x8 cell

64x128

2	3	4	4	3
5	11	17	13	7
11	21	23	27	22
23	99	165	135	85
91	155	133	136	144

Gradient magnitude

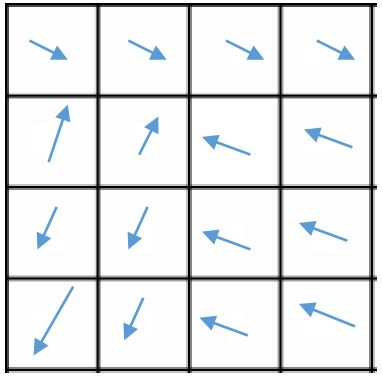
梯度幅值

80	36	5	10	0
37	9	9	179	78
87	136	173	39	102
76	13	1	168	159
120	70	14	150	145

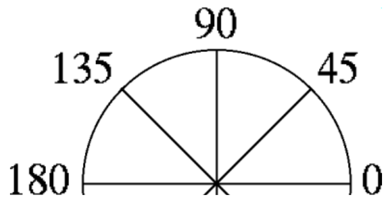
Gradient orientation

梯度方向

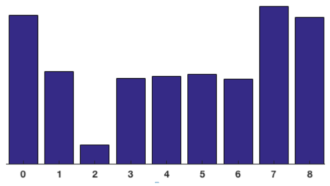
Histograms over k x k pixel cells



Orientation: N bins (N = 6,8, 9, 12, …)



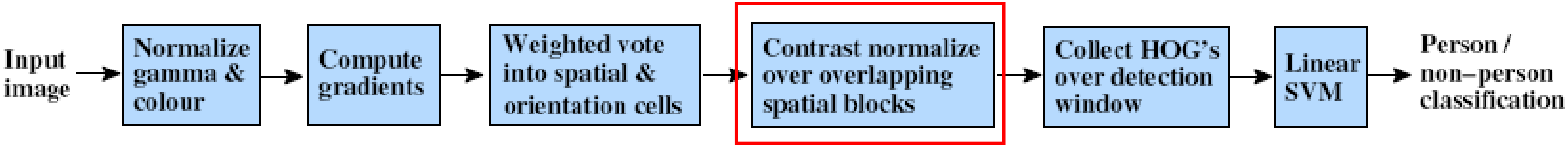
histogram



0，20，…，160

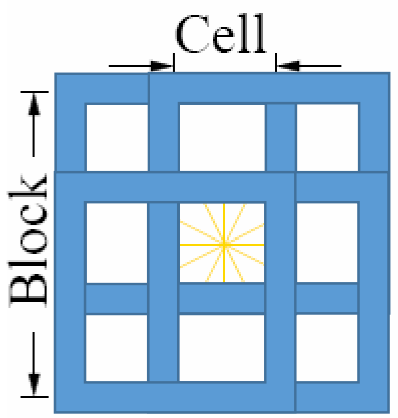
Dalal-Triggs pedestrian detector

HOG Feature extraction



Normalize with respect to surrounding cells

归一化：使特征描述对光照不敏感



$$f = \frac{v}{\sqrt{\|v\|_2^2 + e^2}}$$

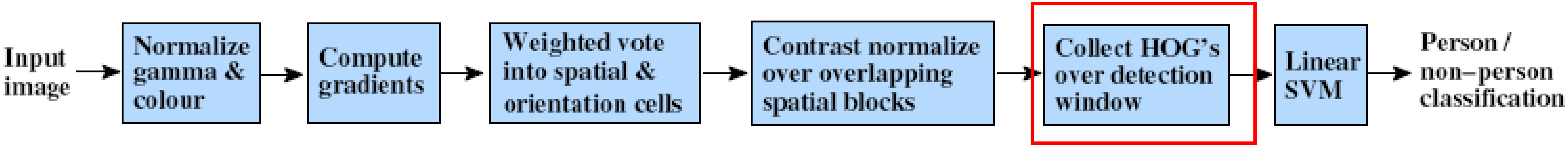
Make descriptor independent of lighting variations

How to normalize?

- Concatenate all cell responses from block into vector.
- Normalize vector.
- Extract responses from cell of interest.
- Do this 4x for each overlapping block.
- Concatenate all the normalized vectors.

Dalal-Triggs pedestrian detector

HOG Feature extraction

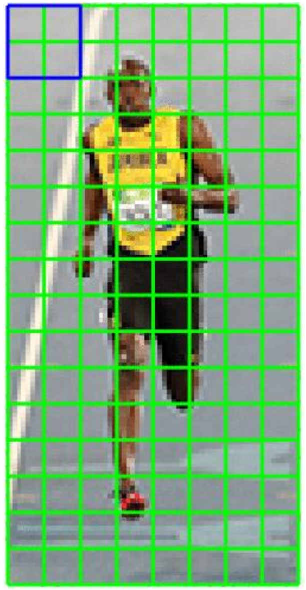


Concatenate the normalized features

16 x 16 block normalization (each block has 2x2 cells)

The cell number is 8 x 16, assume each cell has a vector length of 9

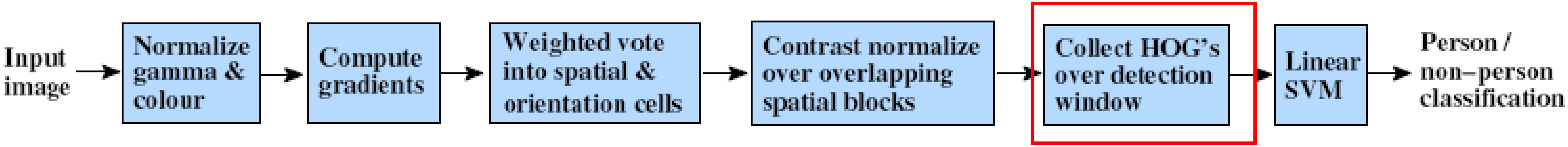
- How many blocks? 7x15
- What is the vector length of each block? 9x4



64x128 pixels

Dalal-Triggs pedestrian detector

HOG Feature extraction



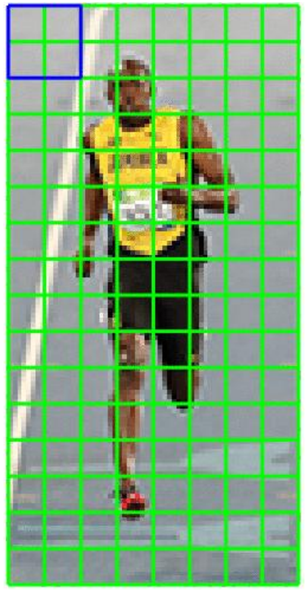
Concatenate the normalized features

16 x 16 block normalization (each block has 2x2 cells)

The cell number is 8 x 16, assume each cell has a vector length of 9

- How many blocks? 7x15
- What is the vector length of each block? 9x4
- What is the feature dimension of the entire region of interest?

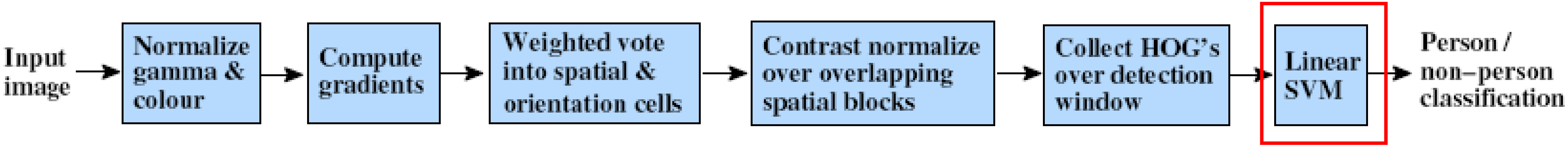
features = 7 x 15 x 9 x 4 = 3780



64x128 pixels

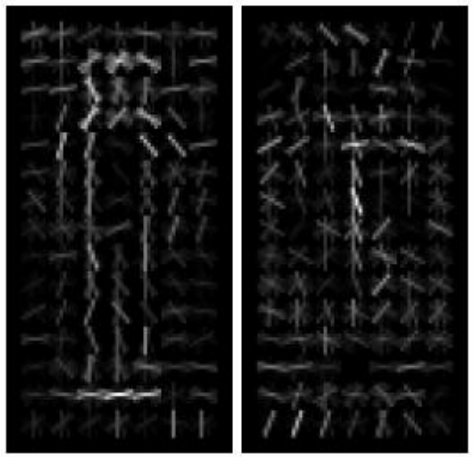
Dalal-Triggs pedestrian detector

Linear classification by SVM

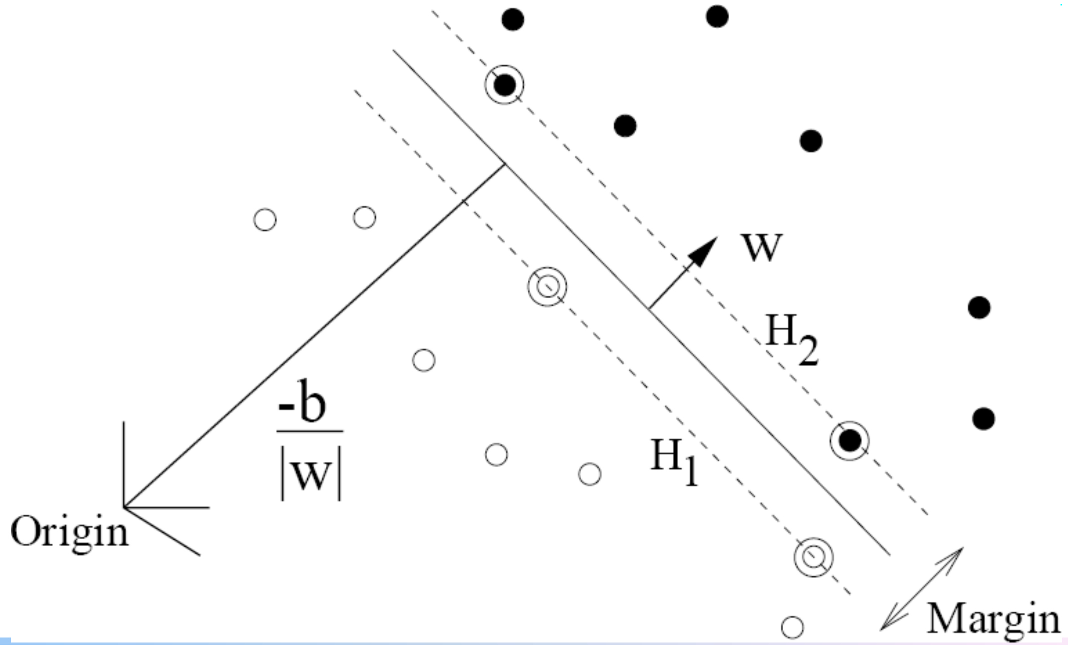


Support vector machine 支持向量机

Training data classes



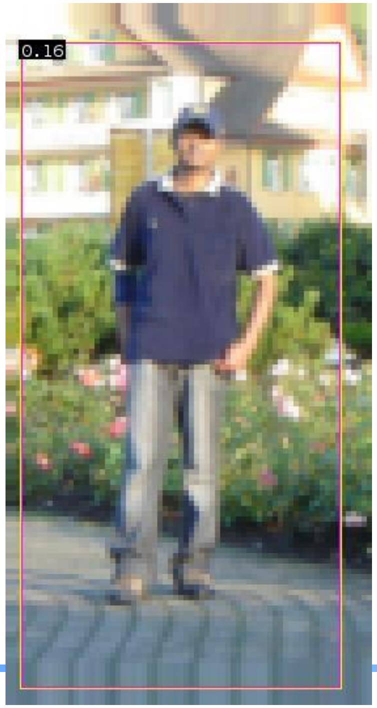
pos w neg w



Dalal-Triggs pedestrian detector

Linear classification by SVM

Input image → Normalize gamma & colour → Compute gradients → Weighted vote into spatial & orientation cells → Contrast normalize over overlapping spatial blocks → Collect HOG's over detection window → Linear SVM → Person / non–person classification



$$0.16 = w^T x - b$$

$$sign(0.16) = 1$$

$=>$ pedestrian

Strengths/Weaknesses of Statistical Template Approach

Strengths

- Works very well for non-deformable objects with canonical orientations: faces, cars, pedestrians
- Fast detection

Weaknesses

- Not so well for highly deformable objects or “stuff”
- Not robust to occlusion
- Requires lots of training data

Content

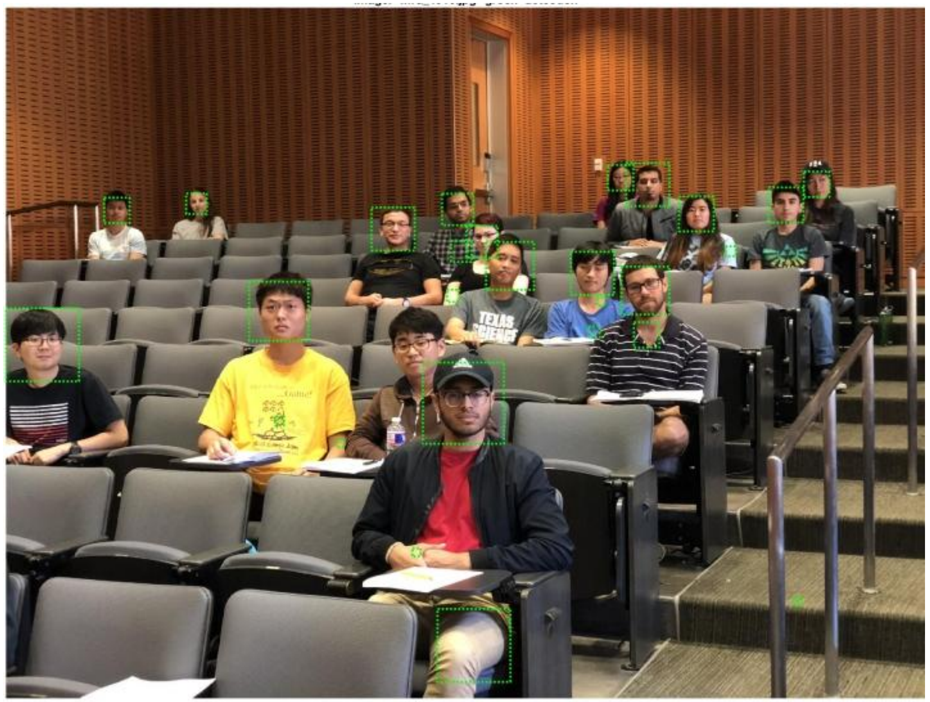
- Object detection basics 目标检测基础
  - What is object detection
  - General process
  - Evaluation
- Pedestrian detection 行人检测
  - Dalal-Triggs method
  - Histogram of oriented gradients
  - Learning with SVM
- Face detection 人脸检测
  - Viola-Jones
  - Boosting for learning

Examples of Face Detection

- Classify a sliding window into face or not face

Face

Not face



Multi-scale method performs better

Challenges of face detection

Sliding window = tens of thousands of location/scale evaluations

- One megapixel image has ~10^6 pixels, and a comparable number of candidate face locations

Faces are rare: 0–10 per image

- For computational efficiency, spend as little time as possible on the non-face windows. 尽早去除非人脸区域
- For 1 Mpix, to avoid having a false positive in every image, our false positive rate has to be less than 10^{-6} 保持低的假阳性率

The Viola/Jones Face Detector

Sliding Window Face Detection with Viola-Jones

Rapid object detection using a boosted cascade of simple features
P Viola, M Jones - … on computer vision and pattern recognition …, 2001 - ieeexplore.ieee.org
This paper describes a machine learning approach for visual object detection which is capable of processing images extremely rapidly and achieving high detection rates. This work is …
☆ 保存 99 引用 被引用次数：27357 相关文章 所有 99 个版本

Robust real-time face detection
P Viola, MJ Jones - International journal of computer vision, 2004 - Springer
… **face detection** framework that is capable of processing images extremely rapidly while achieving high **detection** … allows the features used by our **detector** to be computed very quickly. …
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P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features.* CVPR 2001.

Citation > 27k

P. Viola and M. Jones. *Robust real-time face detection.* IJCV 57(2), 2004.

The Viola/Jones Face Detector

A seminal approach to real-time object detection

Training is slow, but detection is very fast

Key ideas:　积分图、自提升算法、分类器级联

1. *Integral images* for fast feature evaluation

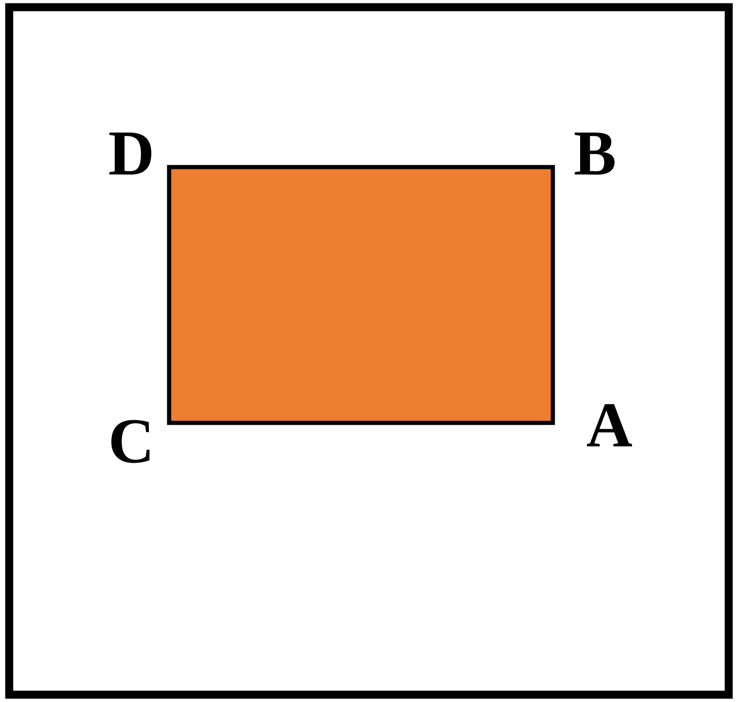
2. *Boosting* for feature selection

3. *Attentional cascade* for fast non-face window rejection

1. Integral images for fast feature evaluation

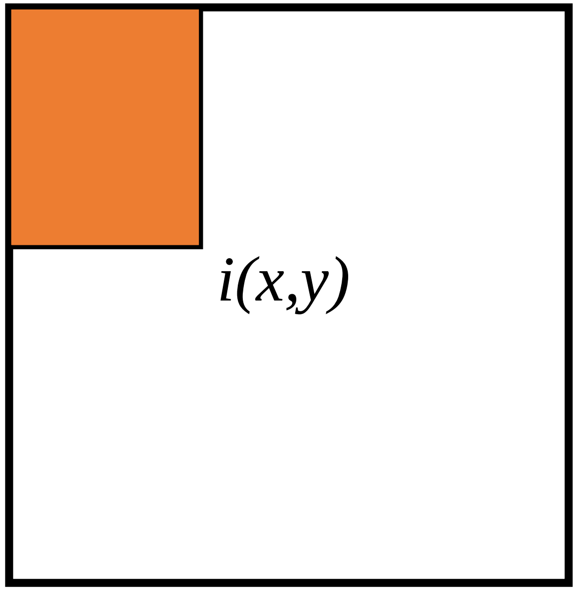
How to calculate the sum of intensity in a rectangle in a fast speed?

如何快速计算矩形区域内像素的灰度之和?



1. Integral images for fast feature evaluation

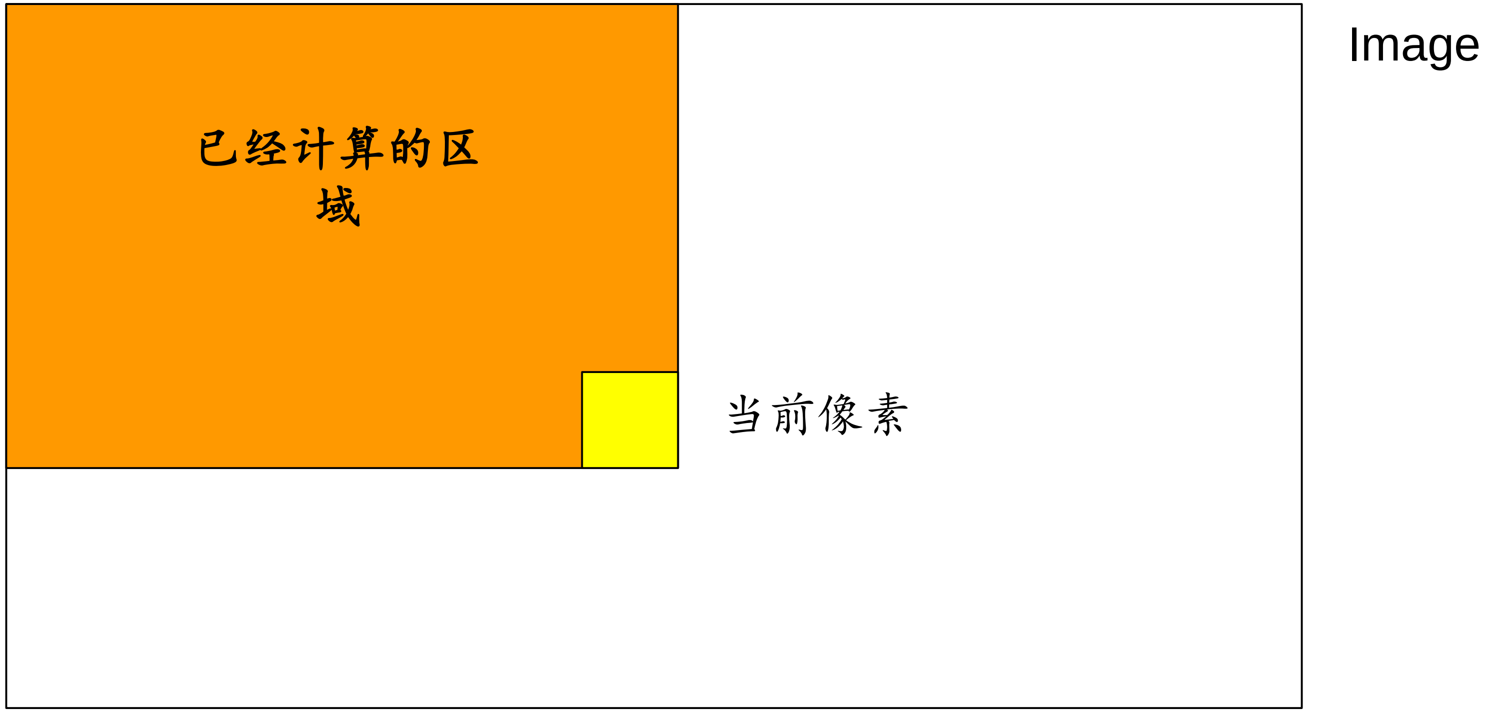
- 积分图（*integral image)*：对于每一个像素（*x,y*)，计算其左上方的所有像素的和。
- 可以仅通过一次遍历进行快速计算。
- 事先将积分值算好，存在表格中供查询。



$$I_{\Sigma}(x,y) = \sum_{\substack{x' \le x \\ y' \le y}} i(x',y')$$

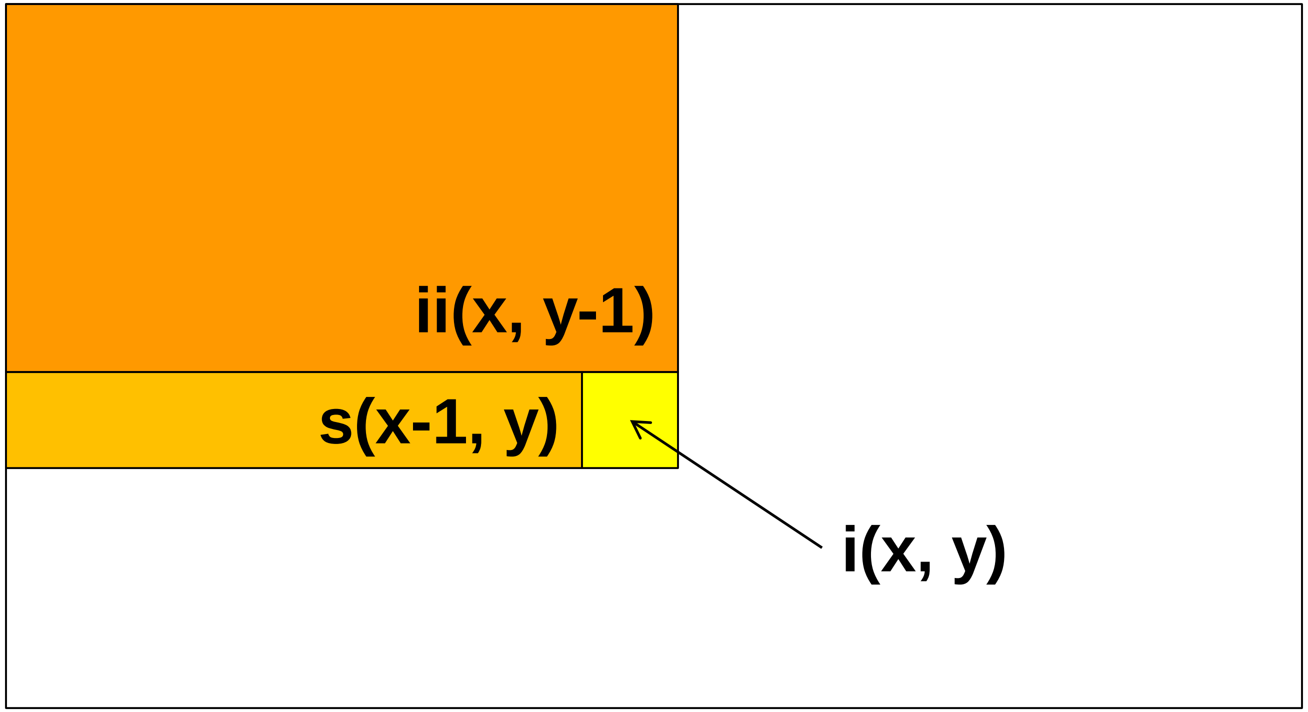
1. Integral images for fast feature evaluation

- 积分图的计算



1. Integral images for fast feature evaluation

- 积分图的计算



- Cumulative row sum:

  $s(x, y) = s(x-1, y) + i(x, y)$

- Integral image

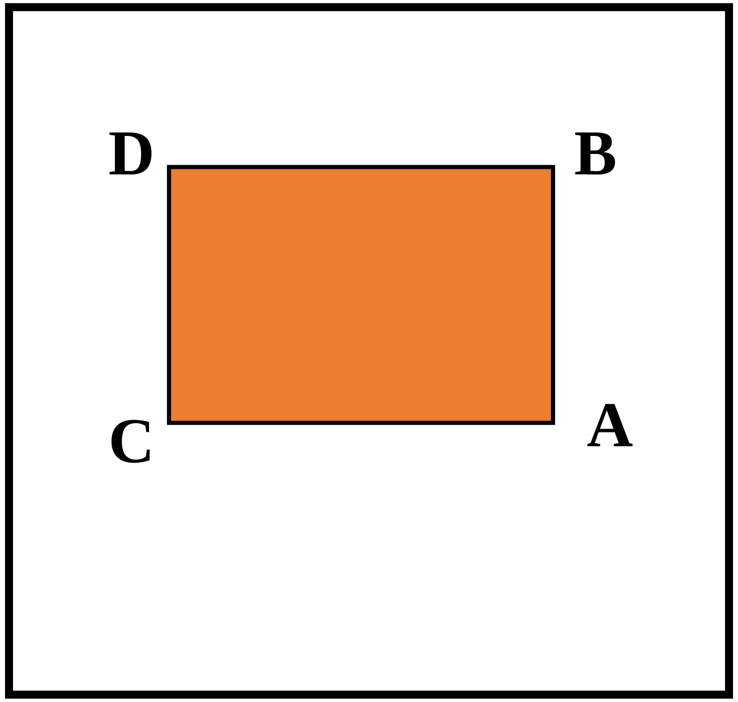
  $ii(x, y) = ii(x, y-1) + s(x, y)$

Python: ii = np.cumsum(i)

1. Integral images for fast feature evaluation

- Computing sum within a rectangle
- Let A,B,C,D be the values of the integral image at the corners of a rectangle
- The sum of original image values within the rectangle can be computed as:

  sum = A – B – C + D



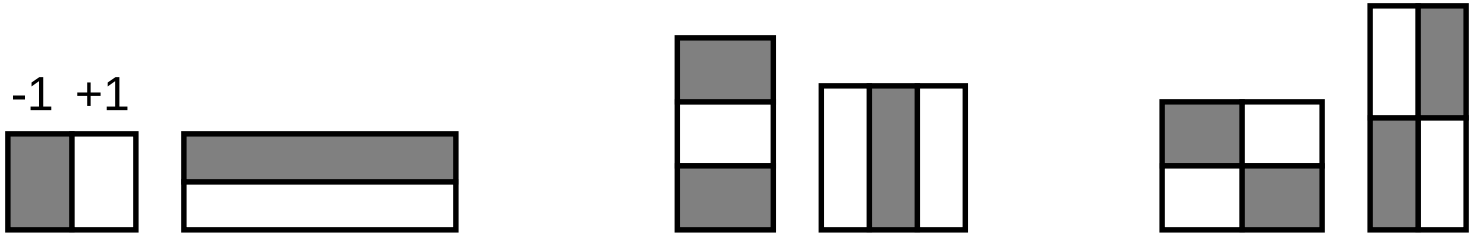
对任意大小的矩形区域，只需要三次加减运算

Only 3 additions are required for any size of rectangle!

1. Integral images for fast feature evaluation

"Haar-like features" 基于积分图的快速特征提取

- Differences of sums of intensity
- Computed at different positions and scales within sliding window

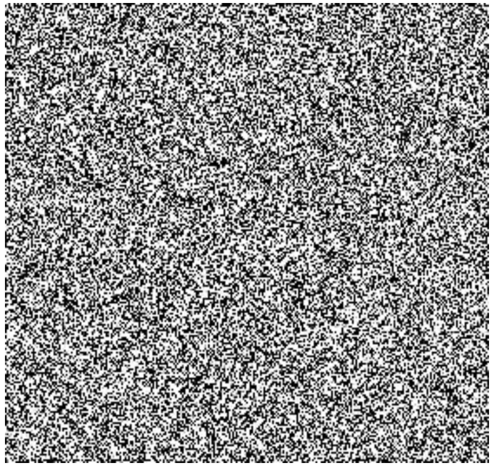


Two-rectangle features Three-rectangle features Etc.

Value = $\sum$*(pixels in white area)* $-\sum$*(pixels in black area)*

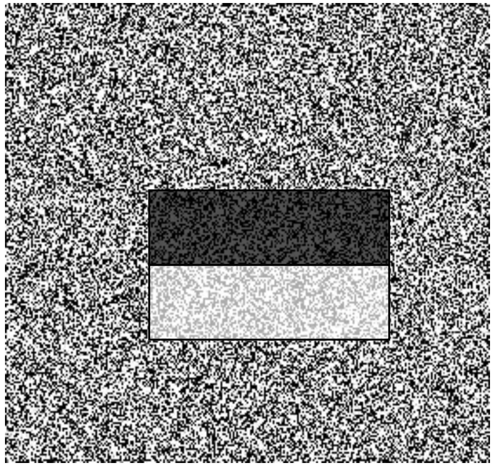
1. Integral images for fast feature evaluation

Example



Source

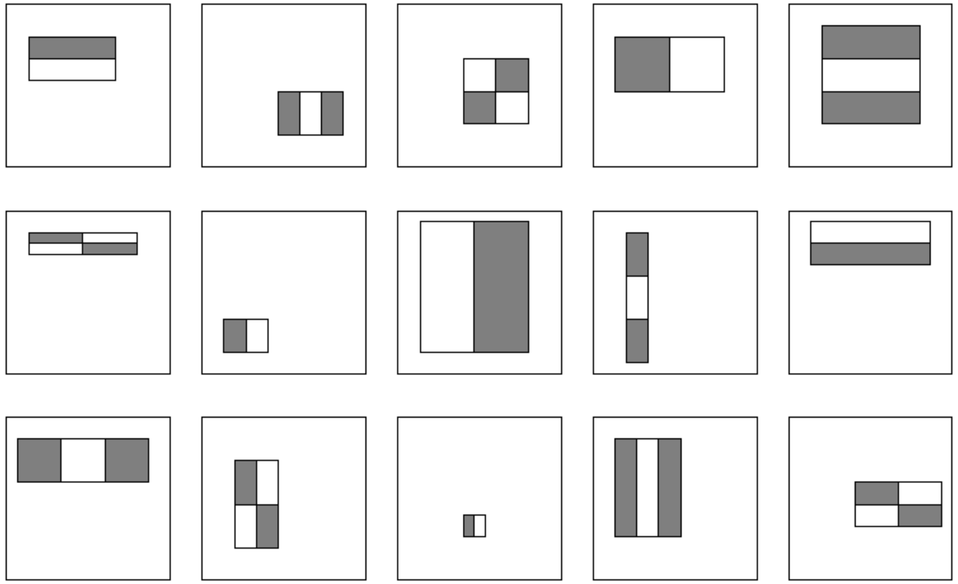
Result



1. Integral images for fast feature evaluation

How many features are there?

For a 24x24 detection region, the number of possible rectangle features is ~160,000!



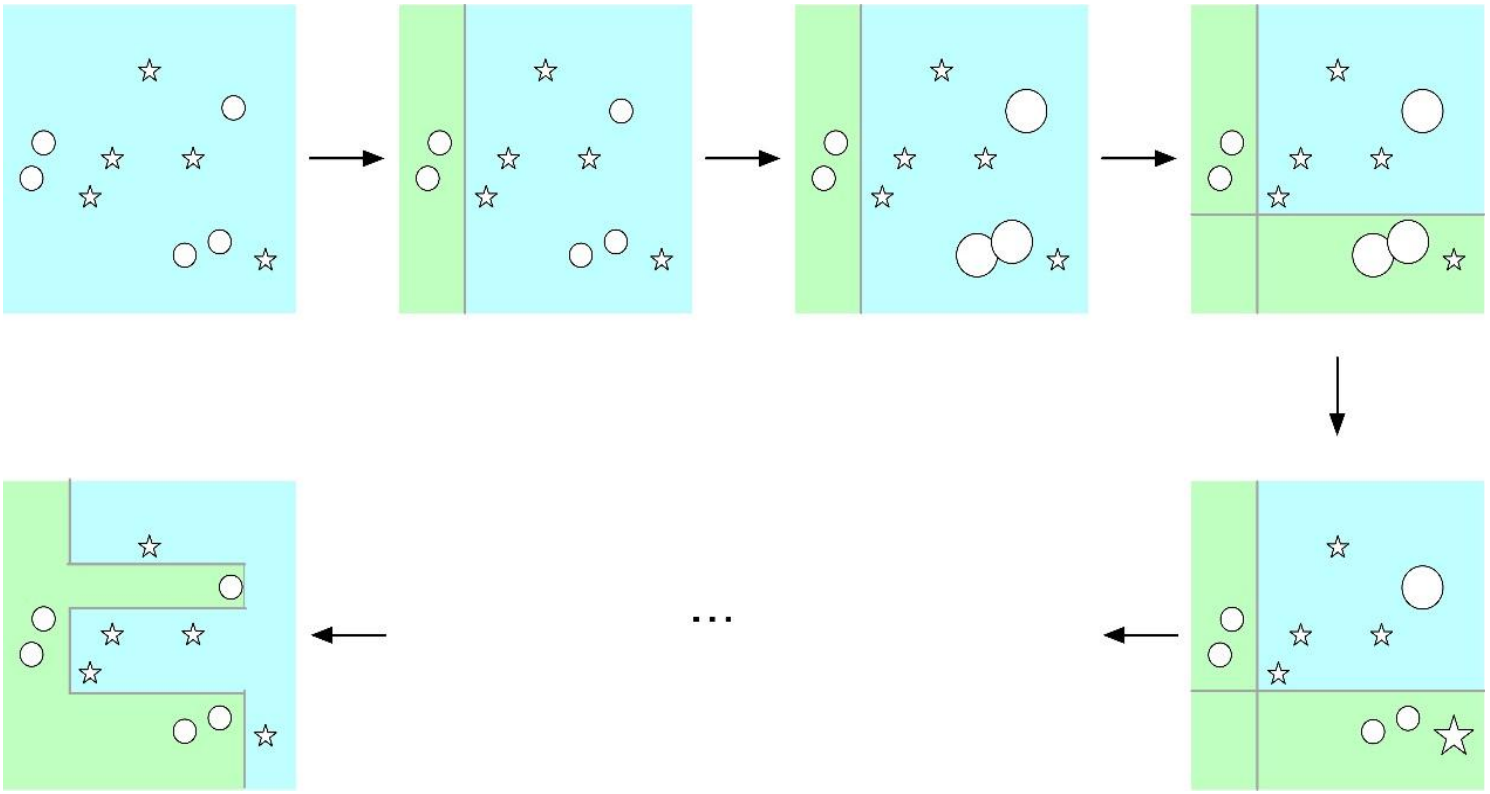
如何使用尽量少的特征进行计算?

- At test time, it is impractical to evaluate the entire feature set.
- Can we learn a ‘strong classifier’ using just a small subset of all possible features?

2. Boosting for feature selection

Adaboosting算法思路

- 由一系列弱分类器级联而成
- 每一个分类器在上一个分类器的基础上进行训练
- 对上一个分类器识别错误的样本赋予更多的权重
- 最终预测结果由这些分类器的预测结果的加权组合而成



2. Boosting for feature selection

Adaboosting算法思路

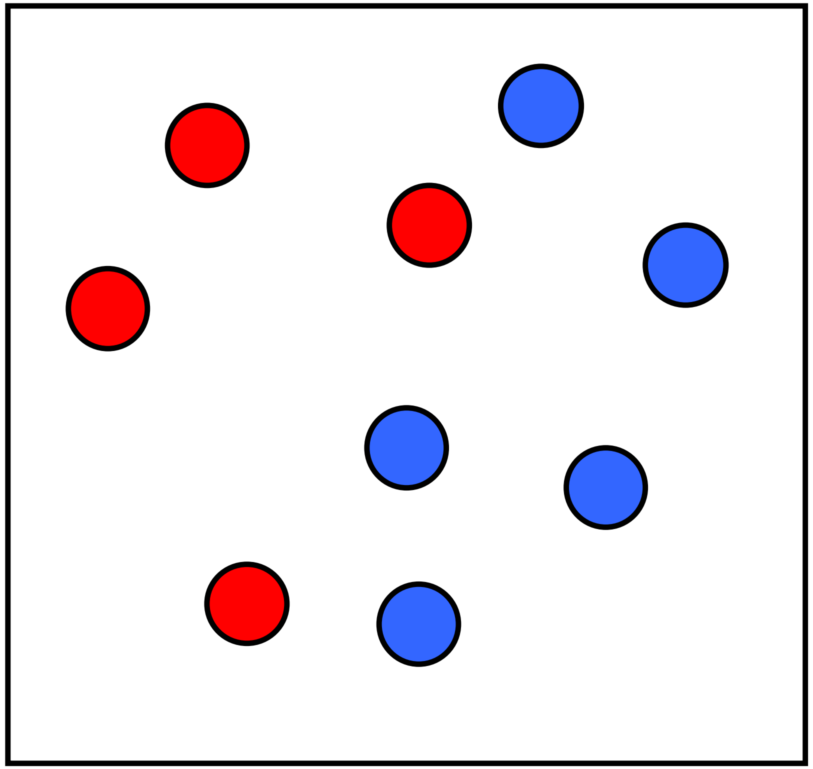
初始阶段，所有样本的权重设为1

对于第1个分类器，样本的权重可写为：

$$D_1 = (w_{11}, \dots, w_{1i}, \dots, w_{1N})$$

其中 $w_{1i} = \frac{1}{N} (1 \le i \le N)$

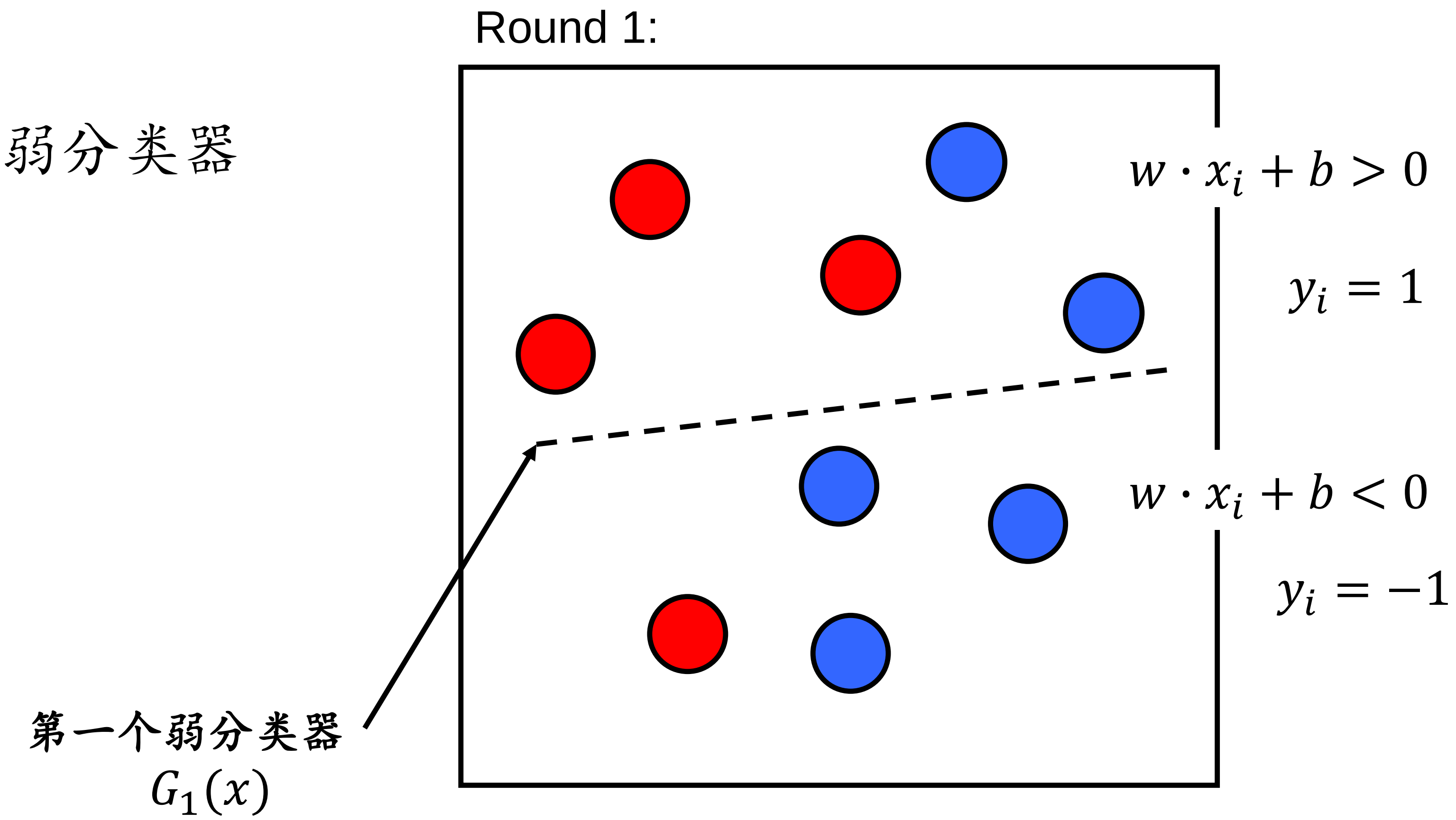
w_{ki}表示第k个分类器中第i个样本的权重



2. Boosting for feature selection

Adaboosting算法思路

在每次迭代中：
- 基于样本的权重求解最佳弱分类器
- 使加权损失函数最小化



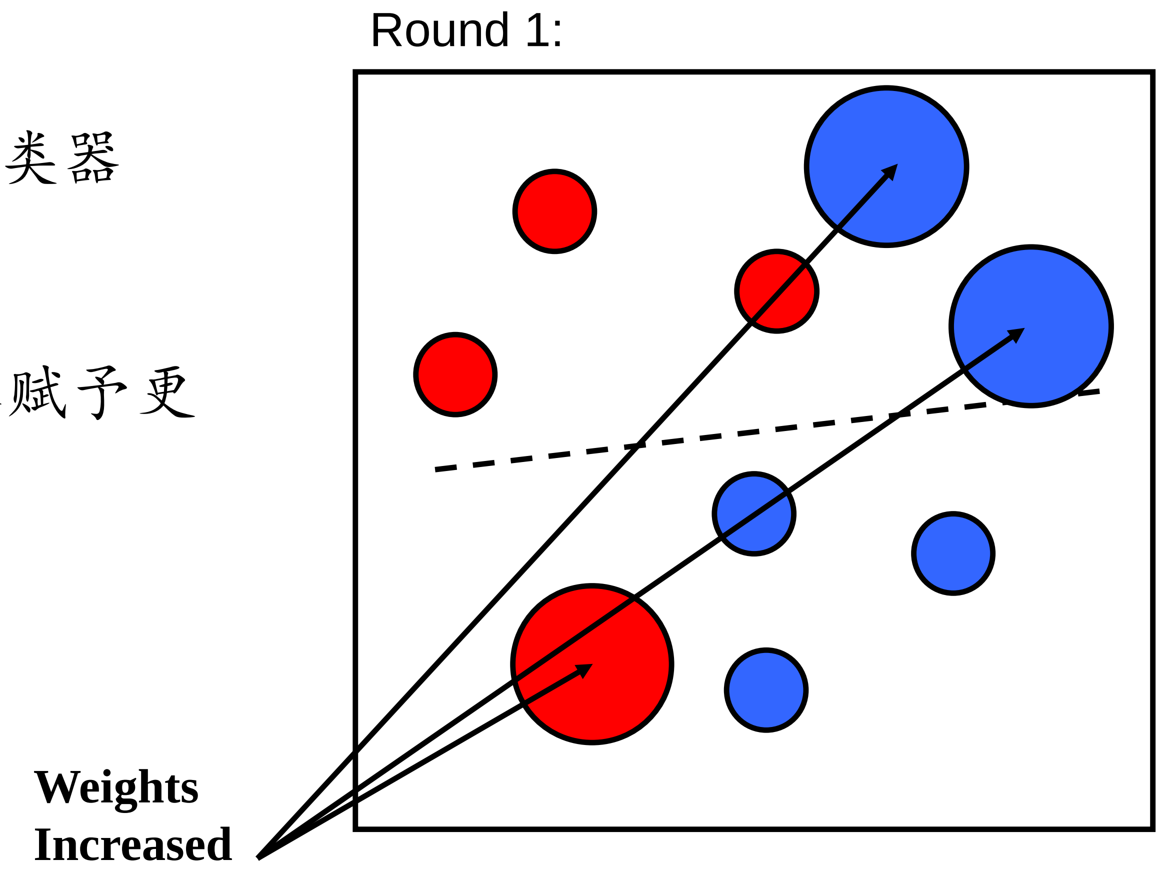
2. Boosting for feature selection

Adaboosting算法思路

在每次迭代中：

- 基于样本的权重求解最佳弱分类器
- 使加权损失函数最小化

- 将当前分类器分类错误的样本赋予更大的权重



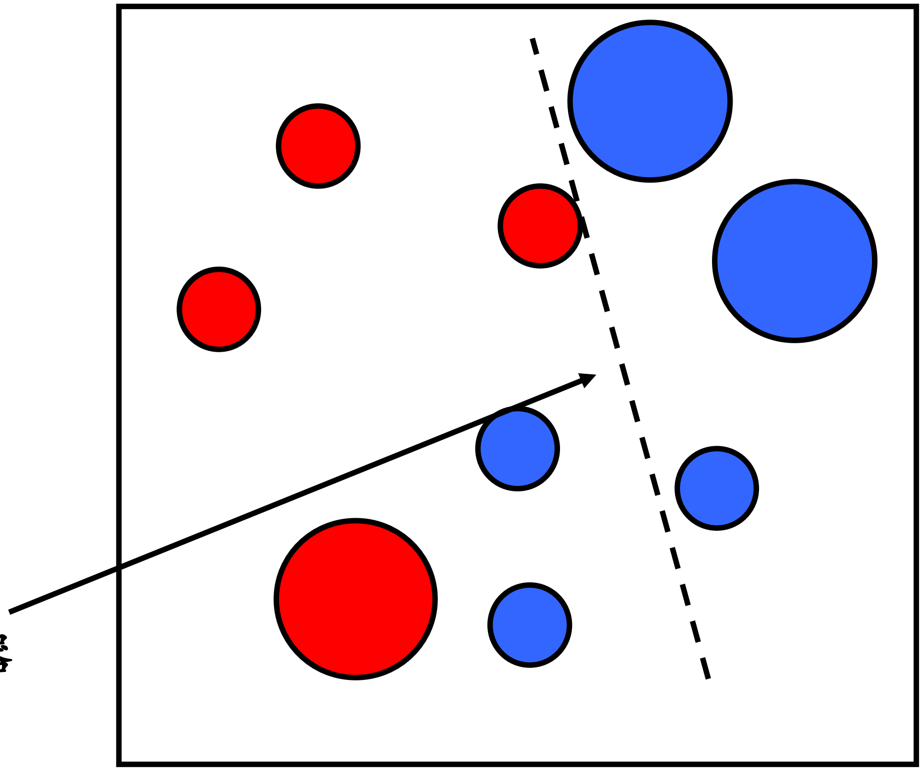
2. Boosting for feature selection

Adaboosting算法思路

在每次迭代中：

- 基于样本的权重求解最佳弱分类器
- 使加权损失函数最小化

Round 2:



第二个弱分类器

$G_2(x)$

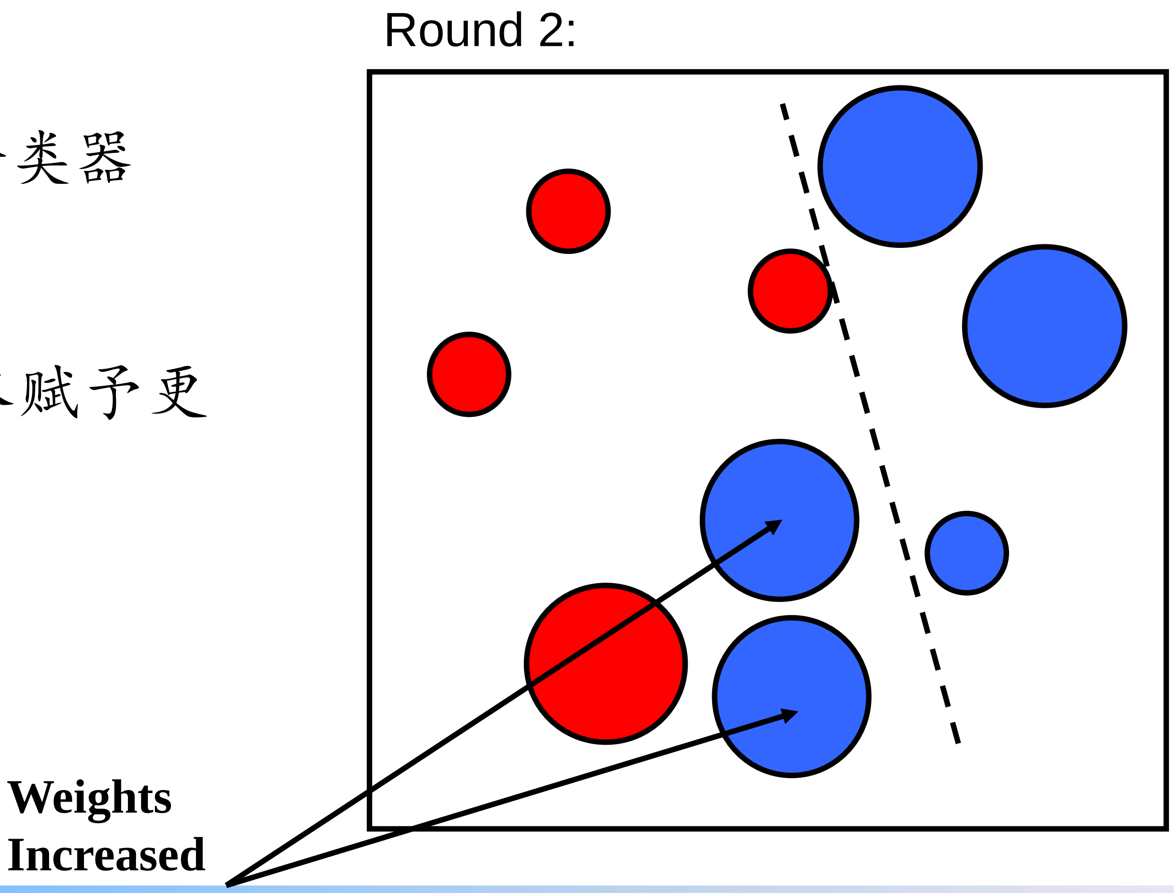
2. Boosting for feature selection

Adaboosting算法思路

在每次迭代中：

- 基于样本的权重求解最佳弱分类器
- 使加权损失函数最小化

- 将当前分类器分类错误的样本赋予更大的权重



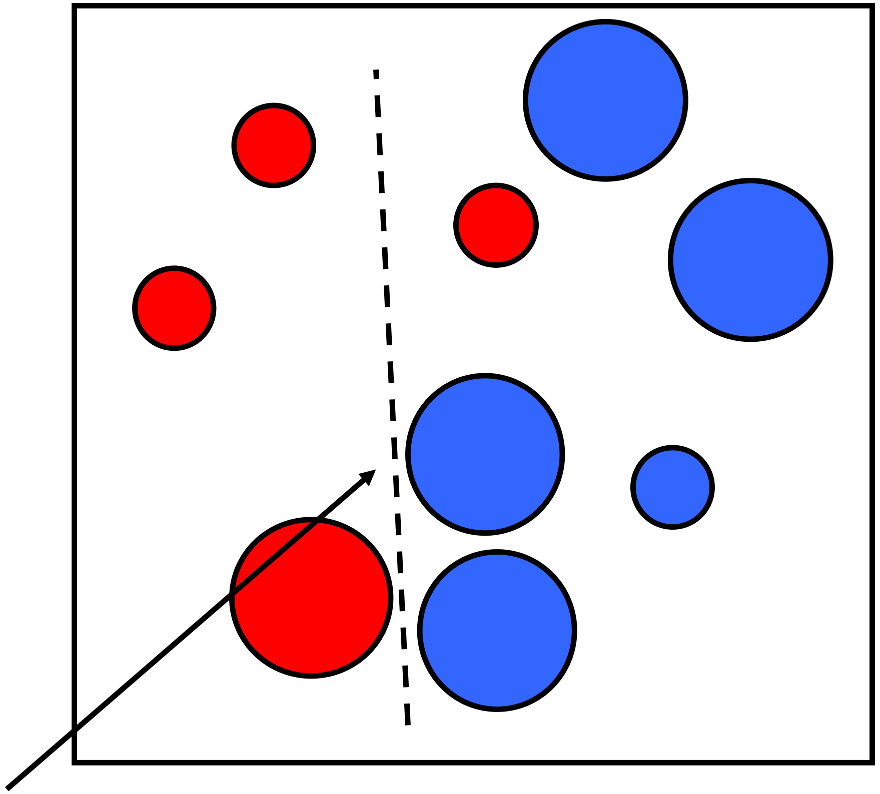
2. Boosting for feature selection

Adaboosting算法思路

在每次迭代中：

- 基于样本的权重求解最佳弱分类器
- 使加权损失函数最小化

Round 3:



第三个弱分类器

$G_3(x)$

2. Boosting for feature selection

Adaboosting算法思路

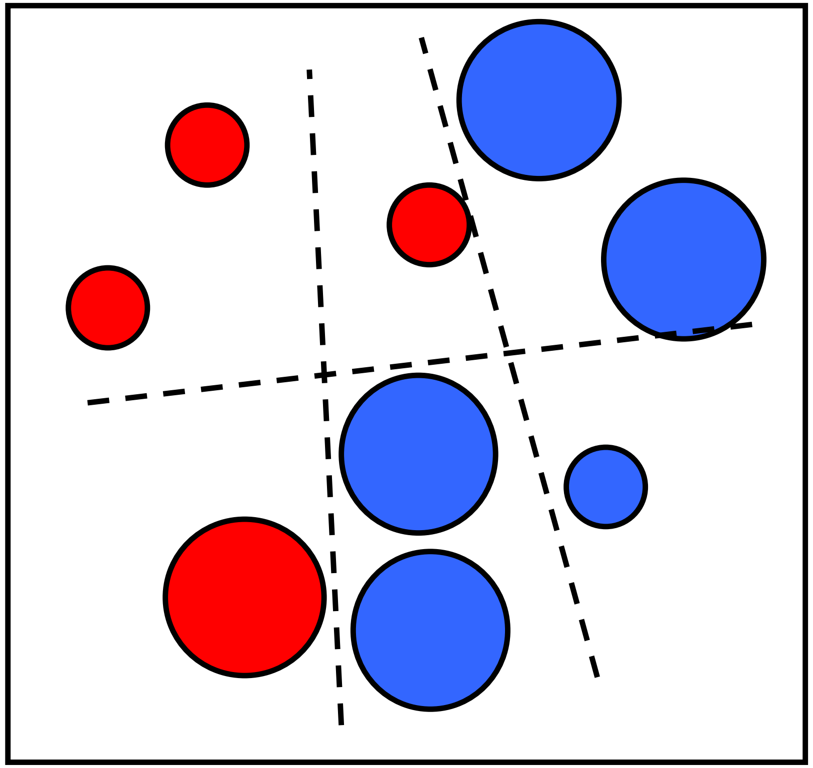
最终的分类器$f(x)$定义为

- 各个弱分类器$G_m(x)$的线性组合
- 每个弱分类器的系数α_m与其精度成反比

$$f(x) = \sum_{m=1}^{M} \alpha_m G_m(x)$$

强分类器　　　　第m个弱分类器

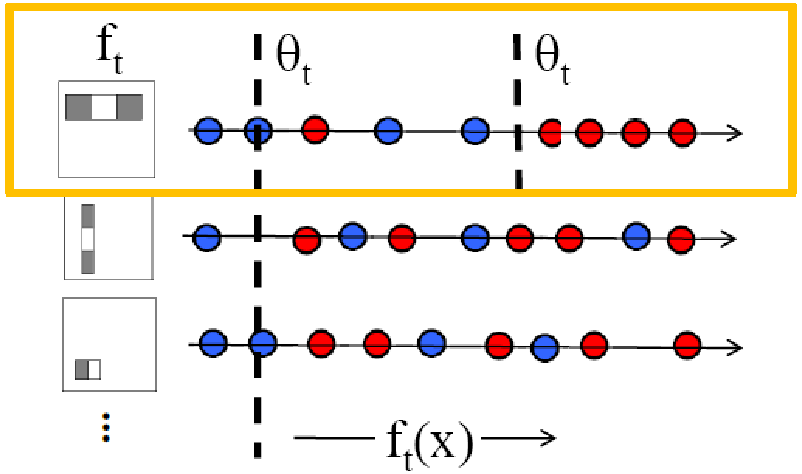
Round 3:



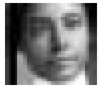
2. Boosting for feature selection

Boosting　　选择单个特征构建弱分类器

Want to select the single rectangle feature and threshold that best separates positive (faces) and negative (non-faces) training examples, in terms of *weighted* error



Outputs of a possible rectangle feature on faces and non-faces.

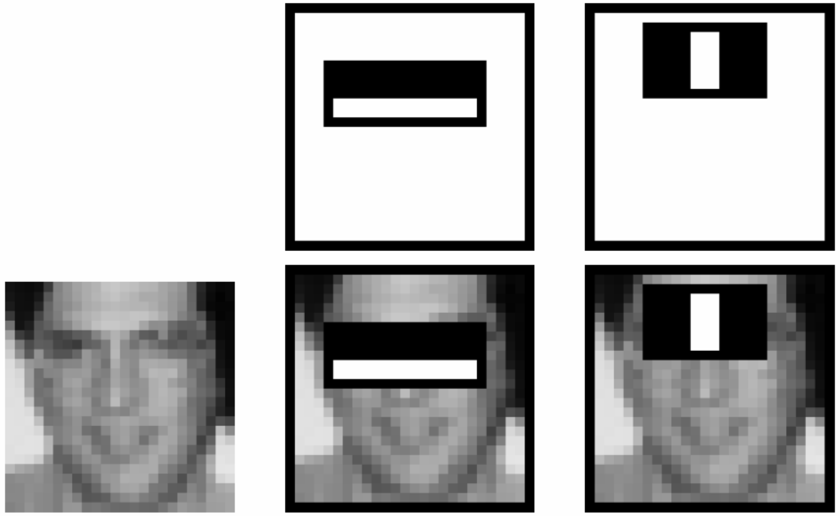


$$h_t(x) = \begin{cases} +1 & \text{if } f_t(x) > \theta_t \\ -1 & \text{otherwise} \end{cases}$$

For next round, reweight the examples according to errors, choose another filter/threshold combo.

2. Boosting for feature selection

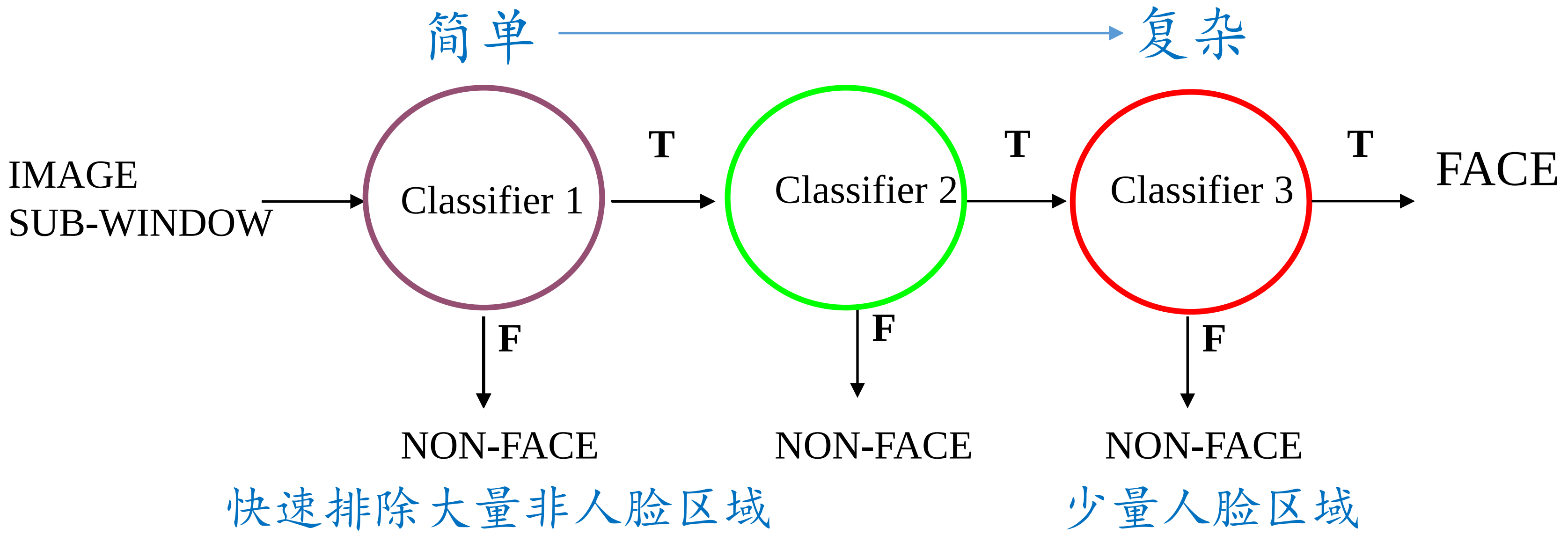
First two features selected by boosting:



This feature combination can yield 100% recall and 50% false positive rate

2. Boosting for feature selection

Cascade for Fast Detection 通过级联实现快速检测



Fast classifiers early in cascade which reject many negative examples but detect almost all positive examples.

Slow classifiers later, but most examples don’t get there.

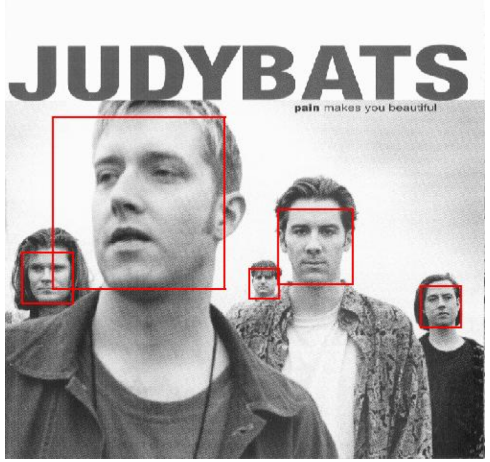
Viola-Jones details

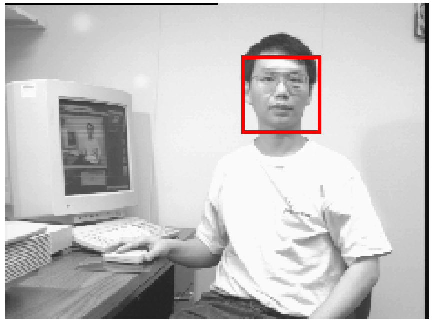
- 38 stages with 1, 10, 25, 50 … features
  - 6061 total used out of 180K candidates
  - 10 features evaluated on average
- Training Examples
  - 4916 positive examples
  - 10000 negative examples collected after each stage
- Scanning
  - Scale detector rather than image
  - Scale steps = 1.25 (factor between two consecutive scales)
  - Translation 1*scale (# pixels between two consecutive windows)
- Non-max suppression: average coordinates of overlapping boxes
- Train 3 classifiers and take vote

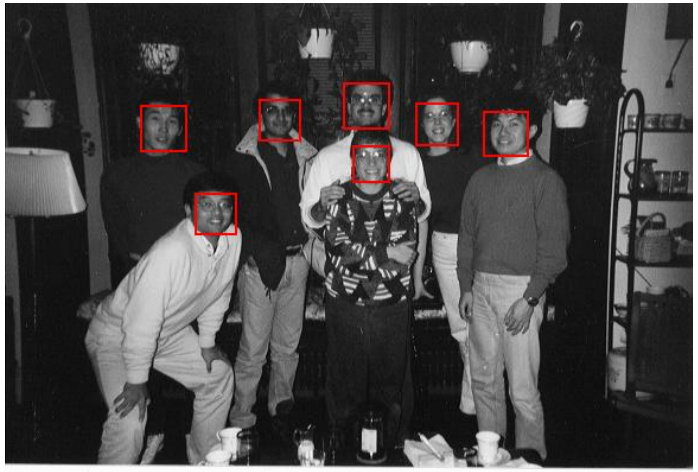
Viola-Jones details

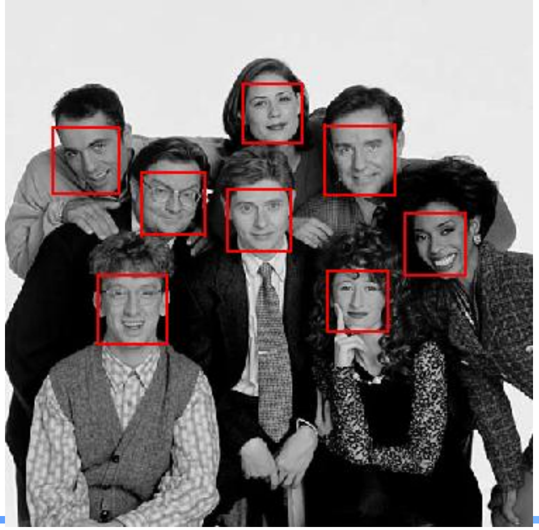
- System performance
- Training time: “weeks” on 466 MHz Sun workstation
- 38 cascade layers, total of 6061 features
- Average of 10 features evaluated per window on test set
- “On a 700 Mhz Pentium III processor, the face detector can process a 384 by 288 pixel image in about 0.067 seconds”
  - 15 Hz
  - 15 times faster than previous detector of comparable accuracy (Rowley et al., 1998)

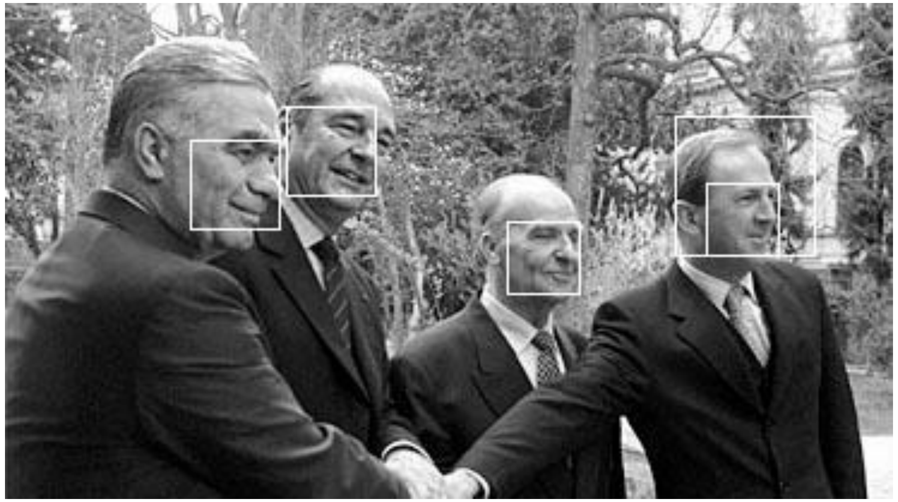
Output of Face Detector on Test Images











小结

- 目标检测基本流程
- Dalal-Triggs 行人检测
- Viola/Jones 人脸检测
  - Rectangle features
  - Integral images for fast computation
  - Boosting for feature selection
  - Attentional cascade for fast rejection of negative windows